FILIPINO TEXT TO MODERN BAYBAYIN SCRIPT IN IMAGE FORM USING N-GRAM & MODERN BAYBAYIN SCRIPT IN IMAGE FORM TO FILIPINO TEXT USING CONVOLUTIONAL NEURAL NETWORK

A Thesis
Presented to the Faculty of the
College of Computer and Information Sciences
Polytechnic University of the Philippines
Sta. Mesa Manila

In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science in Computer Science

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May 2019

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ACKNOWLEDGEMENTS

We would like to acknowledge and express our deepest gratitude to the following for their support and contributions which have been beneficial in developing our study entitled Filipino Text to Modern Baybayin Script in Image Form Using N-Gram & Modern Baybayin Script in Image Form to Filipino Text Using Convolutional Neural Network.

First of all, we would like to give the utmost gratefulness to the Almighty Lord for providing and giving us wisdom, strength, and knowledge needed to accomplish this study. And for His guidance, that allowed us to surpass all the problems encountered in this research.

To our responsible, hardworking, and very considerate thesis adviser, Dr. Aleta C. Fabregas, for guiding us from the beginning, and for motivating us until the end, we are very grateful. Also for being patient for our mistakes.

To the brilliant members of the panel, for their opinions which allowed us to improve and develop the study and by providing their insight.

To our kind and accommodating expert, Mr. Nilo T. Boquiren a Teacher 1 of Panghulo National High School, for his participation in our study.

To our colleagues from PUP-BSCS for providing us their insights. Especially to those who cooperated with us during the implementation of our system.

And finally, to our families who have always been by our side all throughout this journey, for their unconditional love and support.

A.B.A

G.D.N.D

J.G.E

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CERTIFICATION OF ORIGINALITY

This is to certify that the research work presented in this thesis, FILIPINO TEXT TO MODERN BAYBAYIN IN IMAGE FORM USING N-GRAM AND MODERN BAYBAYIN SCRIPT IN IMAGE FORM TO FILIPINO TEXT USING CONVOLUTIONAL NEURAL NETWORK for the degree Bachelor of Science in Computer Science at the Polytechnic University of the Philippines embodies the result of original and scholarly work carried out by the undersigned. This thesis does not contain words or ideas taken from published sources or written works that have been accepted as basis for the award of the degree from any other higher education, institution, except where proper referencing and acknowledgement of work.

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ABSTRACT

Title : Filipino Text to Modern Baybayin Script in Image Form Using N-Gram & Modern Baybayin Script in Image Form to Filipino Text Using Convolutional Neural Network

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This study, Filipino Text to Modern Baybayin Script in Image Form Using N-Gram & Modern Baybayin Script in Image Form to Filipino Text Using Convolutional Neural Network aims to translate Filipino text to Baybayin Script Image and vice versa and help those people who are into this field. The study can translate Filipino Text by syllables, by word and by sentence. After conducting an experiment, the Filipino text to Modern Baybayin translation resulted to 95% accuracy, the Filipino text syllable to Modern Baybayin translation resulted to 98.6% while the Modern Baybayin Script to Filipino text reached 72% of accuracy and Modern Baybayin Script to Filipino text syllable reached 91% of accuracy.

Keywords: Bachelor of Science in Computer Science, Filipino Text, Kudlit, Modern Baybayin, N-Gram, Polytechnic University of the Philippinesm, Script.

Table of Contents

Title Page		**i**
Signature		**ii**
Acknowledgements		**iii**
Certification of Originality		**iv**
Abstract		**v**
Table of Contents		**vi**
List of Tables		**vii**
List of Figures		**x**
1	**The Problem and Its Setting**	**1**
	Introduction	1
	Theoretical Framework	4
	Conceptual Framework	6
	Statement of the Problem	9
	Scope and Limitation of Study	9
	Significance of Study	10
	Definition of Terms	11
2	**Review of Literature and Studies**	**14**
	Related Literature	14
	Related Studies	25
	Synthesis of The Study	33
3	**Methodology**	**40**
	Research Design	40
	Sources of Data	40
	Research Instrumentation	42
	Development Details	45
	Research Instrument	47
	Data Generation/Gathering Procedure	48

Statistical Treatment of Data 48

4 Results and Discussion 50

5 Summary of Findings, Conclusions and Reccomendations 59

Summary of Findings 59

Conclusion 59

Reccomendations 60

References 62

Appendices 66

Appendix 1: Instrument 66

Appendix 2: Correspondence 93

Appendix 3: Certification of Editing 94

Appendix 4: Biographical Statement 95

Appendix 5: Implementation Report 97

LIST OF TABLES

Number	Title	Page
1	**Different forms of Filipino syllables and examples**	26
2	**Synthesis Matrix for Text to Image Translation**	36
3	**Synthesis Matrix for Image to Text Translation**	38
4	**Result of the translation of Filipino Textto Modern Baybayin Script**	51
5	**Result Record of Filipino Text Syllable to Modern Baybayin Script**	51
6	**Result Record of Modern Baybayin Script to Filipino Text Translation**	52
7	**Result Record of Modern Baybayin Script Syllable to Filipino Text**	53
8	**Result Record of Filipino Text to Modern Baybayin Script Translation**	53
9	**Result Record of Filipino Text Syllable to Modern Baybayin Script Translation**	54
10	**Result Record of Modern Baybayin Script to Filipino Text Translation**	54
11	**Result Record of Modern Baybayin Script to Filipino Text Translation**	55
12	**Summary of the Results of the System in terms of Modern Baybayin Script into Filipino Text**	56
13	**Summary of the Results of the System in terms of Modern Baybayin Script into Filipino Text**	56
14	**Comparison of proposed system to existing system of Pedroza et. Al**	58
15	**Filipino Text to Modern Baybayin Script**	68
16	**Modern Baybayin Script to Filipino text**	76
17	**Filipino Text to Modern Baybayin Script**	80
18	**Modern Baybayin Script to Filipino Text**	86

LIST OF FIGURES

Number	Title	Page
1	**Modern Baybayin Script**	3
2	**Theoretical Framework according to Vichyaloetsiri and**	4
3	**Theoretical Framework according to Patel Et Al entitled Optical**	5
4	**Conceptual Framework of the Study**	6
5	**Conceptual Framework of Filipino Text to Modern Baybayin Script**	7
6	**Conceptual Framework of Modern Baybayin Script to Filipino Text**	8
7	**Baybayincharacter "B" with a kudlit**	18
8	**Baybayin with a Spanish + kudlit**	18
9	**Baybayin with X and backslash**	19
10	**Baybayin represent Be, Bi, Bo, Bu script**	19
11	**N-gram visualization**	20
12	**Array of RGB (Red, Green, Blue) Matrix**	22
13	**Neural Network with Convolutional Neural Network**	23
14	**System Architecture of Text to Image Translation**	42
15	**System Architecture of Image to Text Translation**	43
16	**V Model**	45
17	**First User Interface Window**	100
18	**Filipino Text to Modern Baybayin Script Window**	101
19	**About Baybayin Window**	102
20	**About the Author Window**	102

Chapter 1

THE PROBLEM AND ITS SETTING

1.1 Introduction

The word *baybayin* is a very old Filipino term that refers to the ancient script writing of the Filipinos prior to the Spanish colonization. Erroneously called *alibata,* Baybayin is a member of the Brahmic family used throughout South and Southeast Asia and was said to be used by ancient Filipinos as early as the 13th century (Tan, 2014) long before the Spaniards came into the Philippines.

Originating from the word *baybáy,* Baybayin literally means to "spell". Incorrectly known as "Alibata", baybayin has been a core part of our culture and heritage for centuries. It has been the soul interconnectivity of our ancestors through writing. To spell; rooted from the word Baybay, our ancestors used our very own writing script to sign documents, write poetries, communicate, and write letters to one another. It was a form of communication that we once utilized in our everyday lives and was very well respected by the explorers who came to our islands." (Bayani Art, 2018).

Recently, a House Committee Bill entitled Baybayin Act (House Bil 1022) or the National Writing System Act was passed which seeks to install Baybayin as the "National Writing System of the Philippines" – declaring it so and "providing for its promotion, protection, preservation and conservation". This declaration, consonant with the declared "policy of the state to inculcate, propagate and preserve the cultural heritage and treasures of the Philippines…" is necessary to make Baybayin "a tool for cultural and economic development to create consciousness, respect and pride for the legacies of Filipino cultural history, heritage and our authentic identity" **(Section 2)** (WindaViews, 2018).

There is an existing research of translating Filipino Text to Baybayin Characters using N-Gram. They used the Old Baybayin Characters which is compose of 17 base characters only in translation. They are having a hard time translating Proper nouns because there are some of proper nouns that doesn't have equivalent to Old Baybayin Characters. At the end of their research they recommend to have the translation of proper nouns and to also have bi-directional approach. With this, the researchers proposed a study that would convert Filipino Text into Modern Baybayin Script using N-gram algorithm and a bi-directional translation of Modern Baybayin Script into Filipino Text using Image Processing and Convolutional Neural Networks or CNN. Modern Baybayin Characters is composed of 28 base characters which says that it can translate proper nouns because all the alphabet has equivalent to Modern Baybayin Characters. Through this proposal the researchers aim to develop an accurate system using different computer science approach in hopes of helping the society in spreading the awareness about our ancestral roots of Modern Baybayin.

Figure 1. **Modern Baybayin Script**

Baybayin - Philippine National Writing System

(Modern/Modified Baybayin)

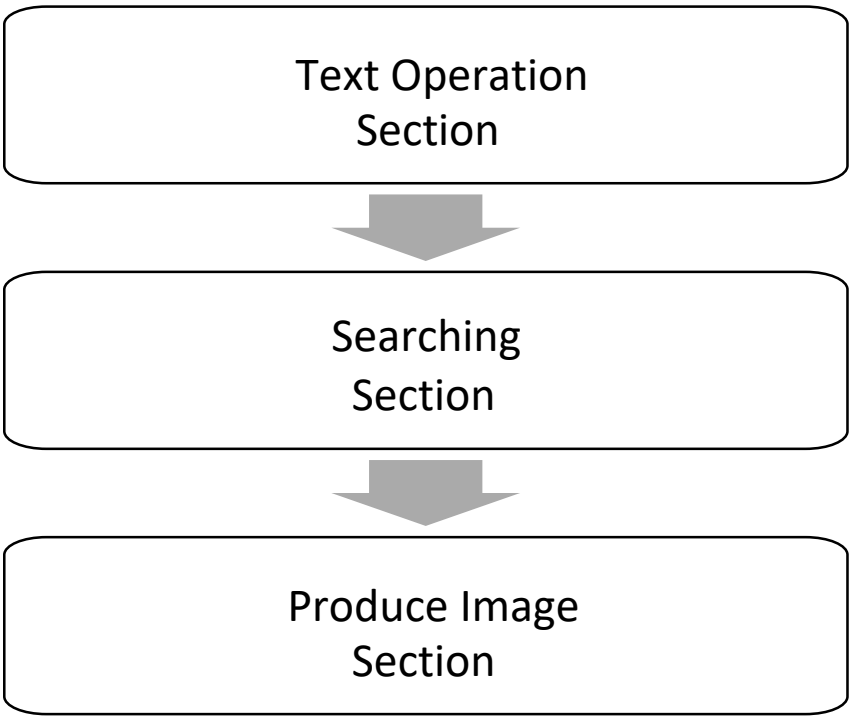
	A =	E =	I =	O =	U=	
Ka =	Ke =	Ki =	Ko =	Ku =	K =	
Ga =	Ge =	Gi =	Go =	Gu =	G =	
Nga=	Nge=	Ngi=	Ngo=	Ngu=	Ng =	
Ca =	Ce =	Ci =	Co =	Cu =	C =	
Ja =	Je =	Ji =	Jo =	Ju =	J =	
Ña =	Ñe =	Ñi =	Ño =	Ñu =	Ñ =	
Ta =	Te =	Ti =	To =	Tu =	T =	
Da =	De =	Di =	Do =	Du =	D =	
Na =	Ne =	Ni =	No =	Nu =	N =	
Pa =	Pe =	Pi =	Po =	Pu =	P =	
Ba =	Be =	Bi =	Bo =	Bu =	B =	
Ma =	Me =	Mi =	Mo =	Mu =	M =	
Ya =	Ye =	Yi =	Yo =	Yu =	Y =	
Ra =	Re =	Ri =	Ro =	Ru =	R =	
La =	Le =	Li =	Lo =	Lu =	L =	
Wa =	We =	Wi =	Wo =	Wu=	W =	
Sa =	Se =	Si =	So =	Su =	S =	
Ha =	He =	Hi =	Ho =	Hu =	H =	
Fa =	Fe =	Fi =	Fo =	Fu =	F =	
Qa =	Qe =	Qi =	Qo =	Qu =	Q =	
Va =	Ve =	Vi =	Vo =	Vu =	V =	
Xa =	Xe =	Xi =	Xo =	Xu =	X =	
Za =	Ze =	Zi =	Zo =	Zu =	Z =	

Kawil " ᜅ " and Pamudpud " ᜉ " are also used as Vowel Killer aside from Sinawali " X "

modernbaybayin.blogspot.com
May 9, 2014

1.2 Theoretical Framework

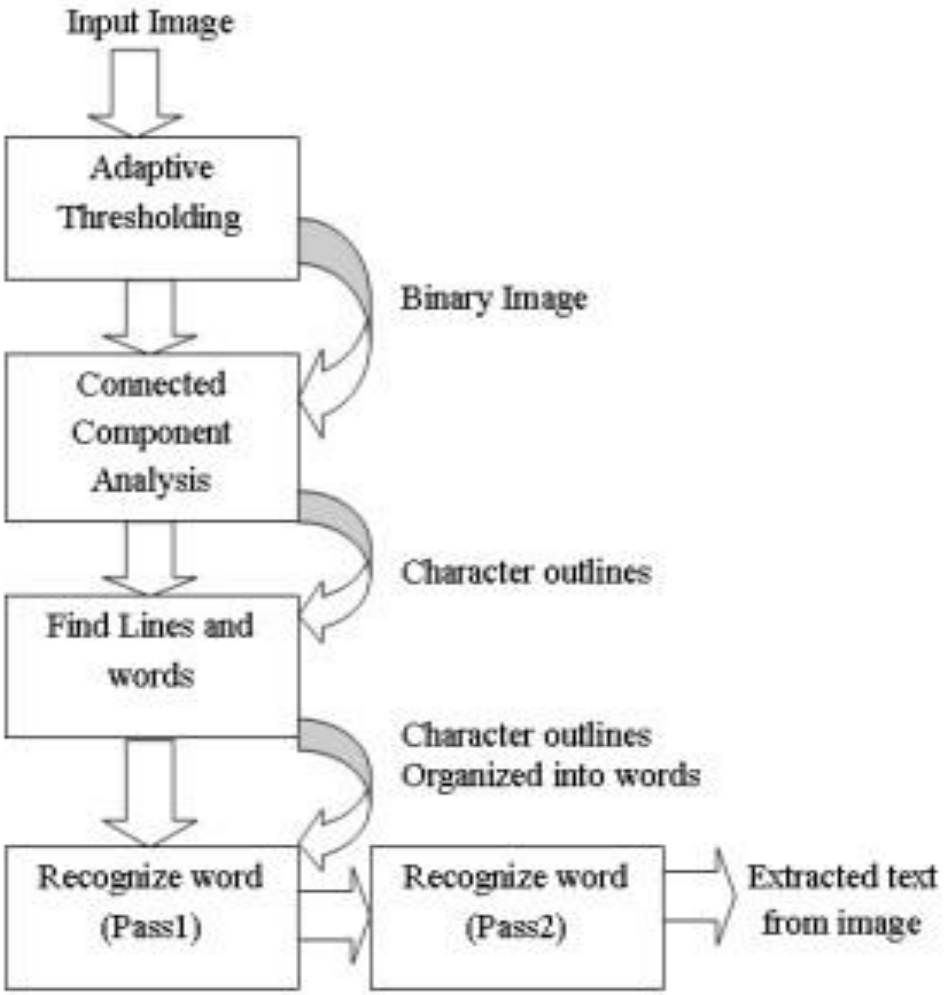
Figure 2. **Theoretical Framework according to Vichyaloetsiri and Wuttidittachotti for translating text into image**



In translating from one language into another writing system this is the theoretical framework that the researcher use. According to research of Vichyaloetsiri and Wuttidittachotti (*International Conference on Computer*, 2017), they present a framework to translate text into a sign language. In text operational section, when users enter a sentence, it will analyze the sentence by the use of natural language processing and it happens in the searching section, then produce a series of image per word that happen in produce image section. In this theory the researcher uses the framework of Vichyaloetsiri and

Wuttidittachotti in conducting a research for Filipino word translate into Modern Baybayin script. By using the n-gram for translatting Filipino words into modern Baybayin, the user enter sentence and analyze the sentence per word and the word will be syllable and it will present a Modern Baybayin alphabet that in image series.

Figure 3. Theoretical Framework according to Patel Et Al entitled Optical Character Recognition by Open Source OCR Tool Tesseract, An Image to Text Translation



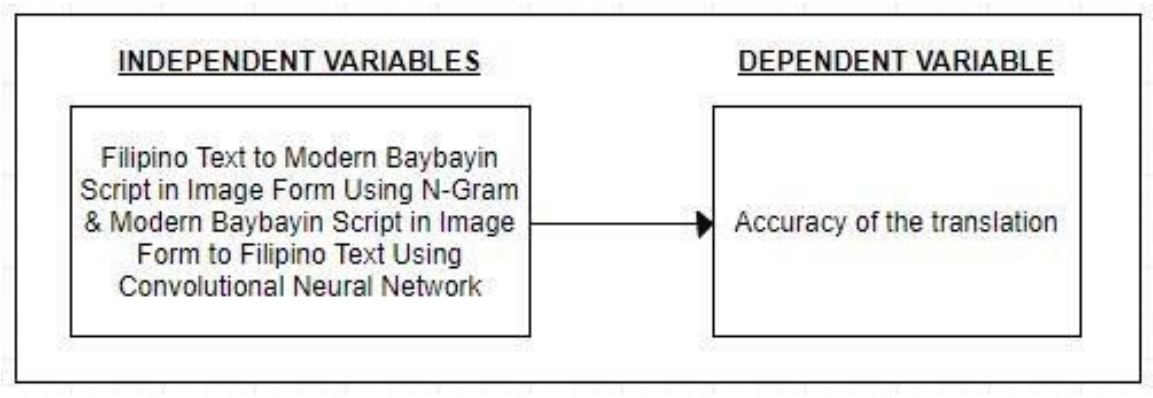
According to research of Patel Et Al entitled Optical Character Recognition by Open Source OCR Tool Tesseract, An Image to Text translation. OCR works in step by step manner as per the block diagram shown in fig. 1. First step is Adaptive Thresholding,

which converts the image into binary images. Next step is connected component analysis, which is used to extract International Journal of Computer Applications (0975 – 8887) Volume 55– No.10, October 2012 51 Fig. 1. Architecture of Tesseract OCR character outlines. This method is very useful because it does the OCR of image with white text and black background. Tesseract was probably first to provide this kind of processing. Then after, the outlines were converted into Blobs. Blobs were organized into text lines, and the lines and regions were analyzed for some fixed area or equivalent text size. Text is divided into words using definite spaces and fuzzy spaces. Recognition of text is then started as two-pass (Patel Et Al., 2012). From this theory the researcher gonna use it for image to text translation from Modern Baybayin to Filipino text, first the image acquire by the system and then the words enter will be divided and then syllabi and check if the Baybayin have the same data in the neurons that the CNN have testing data in and it produce a corresponding text by the Baybayin gathered image.

1.3 Conceptual Framework

1.3.1 Conceptual Framework of The Study

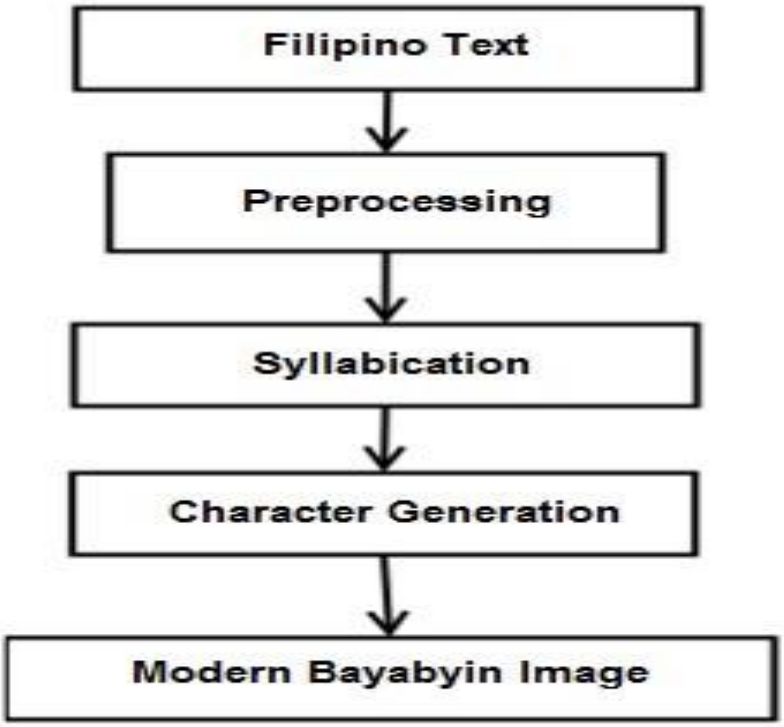
Figure 4. **Conceptual Framework of the Study**



The above figure (Figure 1.6) show the Conceptual Framework of this study, Filipino Text to Modern Baybayin Script in Image Form Using N-Gram Modern Baybayin Script in Image Form to Filipino Text Using Convolutional Neural Network.

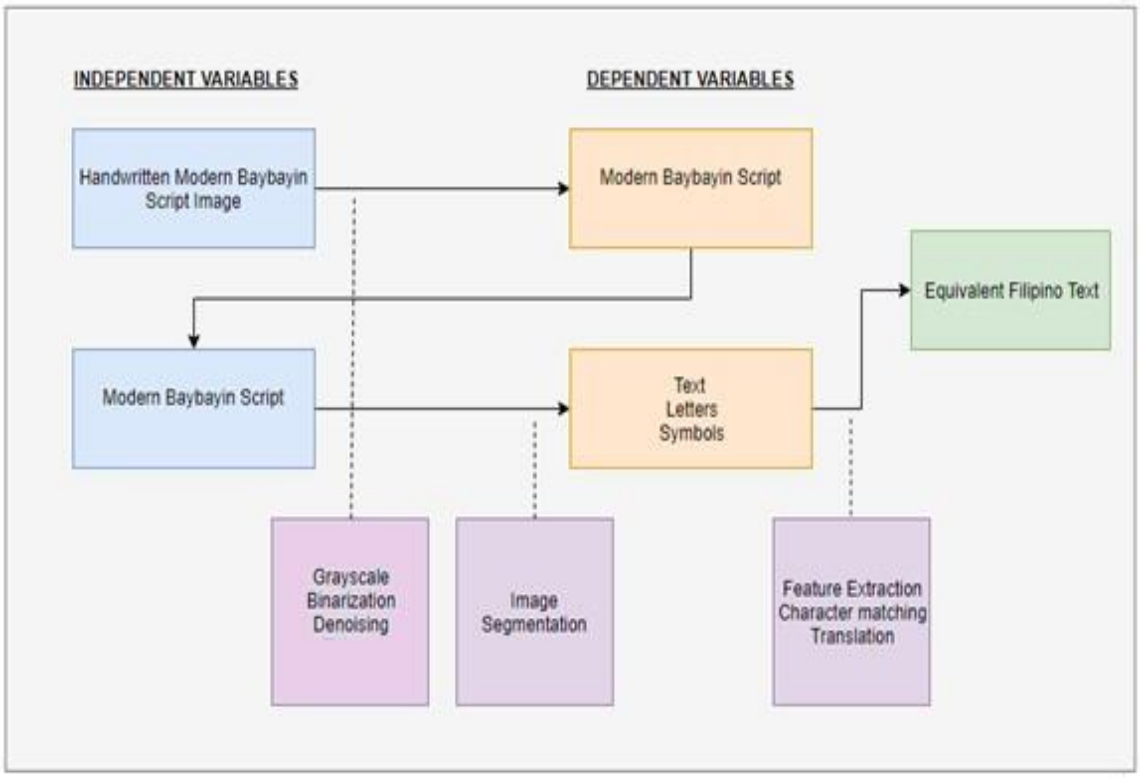
1.3.2 Conceptual Framework of The System

Figure 5. **Conceptual Framework of Filipino Text to Modern Baybayin Script**



For the translation of Filipino Text to Modern Baybayin Script, the user starts with the input of the words to be translated, next one would be the validation of words for filtering, then the system performs the syllabication of the input which would be fed to the character generation for the finallk phase of translating it and displays the corresponding Modern Baybayin image characters.

Figure 6. **Conceptual Framework of Modern Baybayin Script to Filipino Text**



The figure above shows Independent and Dependent variable relationship to each other. The Independent variable used to produce dependent variables using techniques like preprocessing, grayscale, Binarization, denoising and image segmentation. The handwritten image produced modern baybayin script using those techniques being said. And modern baybayin script produced text, letters and symbols and over all to produce the desired output which is Filipino Text.

1.4 Statement of The Problem

This study aims to solve the problem about translating proper nouns of the existing system by Pedroza et. al and to increase the degree of accuracy using convolutional neural network in correctly recognizing modern baybayin characters. In addition, the researchers intend to answer the following questions at the end of the study:

1. What is the degree of accuracy of translation with respect to:
    a. Filipino Text to Modern Baybayin script?
    b. Filipino Text Syllable to Modern Baybayin Script?
    c. Modern Baybayin Script to Filipino Text Syllable?
    d. Modern Baybayin script to Filipino text?
2. What is the significant difference in the translation of the proposed system and the existing system of Pedrozaet. al?

1.5 Scope and Limitation of Study

Scope

The scope of the system would only to translate Filipino Text to Modern Baybayin Script in Image Form and Modern Baybayin Script in Image Form to Filipino Text only.

Limitation

The study would only be limited to Windows based translation of Filipino Text to Modern Baybayin Script in Image Form and Modern Baybayin Script in Image Form to Filipino Text.

- Translation of Filipino Text to Modern BaybayinScipt in Image Form

In Filipino text to Modern Baybayin translation, it will only translate Filipino language. In case the user input other language other than Filipino, it will not translate by the system.

- Translation of Modern Baybayin Script in Image Form to Filipino Text

In Modern baybayin conversion to Filipino text, the input image should be a scanned copy of the handwritten Modern Baybayin script written in a white clean bond paper using black ink only. Lastly the Modern baybayin content to be translated is assumed to be in Filipino language.

1.6 Significance of Study

The findings of this study would be significant and beneficial to the following individual/organizations:

- **Filipino Teachers** – Filipino Teachers who wants to learn and teach Baybayin in the near future would greatly benefit from this research. This will be a very big help to them in teaching their students the ancient way of writing.
- **Society –** Since the learning of Baybayin is very timely nowadays because of the signed House Committee Bill; the society (i.e. LGUs, manufactures, newspapers) may use this research for easing the difficulty of understanding and using Baybayin in their operations.
- **Future Researchers –** This study will also benefit future researchers for the findings of this research may serve as their reference or guide in doing their own study.
- **Students -** The produced system of the study could be used by students in reinforcing their knowledge in the area of Baybayin script.

- **Philologists, Calligraphists & Linguistics –** This study would be beneficial for those people who are interested in the field of ancient writings for this study could greatly aid them not just in learning and preserving Baybayin but also in helping in spreading their knowledge about this particular ancient writing system

- **Historians –** Historians especially those in the field of indigenous writings can use the outputs of this research to aid them in further advancing their works and in preserving this cultural heritage.

1.7 Definition of Terms

The following terms used in this study were defined operationally.

- **Brahmic Family.** or scripts are a ***family*** of abugida or alphabet writing systems.

- **Convolutional neural network** (**CNN**, or **ConvNet**)**.** is a class of deep, feed forward artificial neural networks, most commonly applied to analyzing visual imagery.CNNs use a variation of multilayer perceptrons designed to require minimal preprocessing. They are also known as shift invariant or space invariant artificial neural networks (SIANN), based on their shared-weights architecture and translation in variance characteristics. For example, inimage recognition, they might learn to identify images that contain Modern Baybayin Script/s by analyzing example images that have been manually labeled as "ma" or "na" and using the results to identify Modern Baybayin Script/s in other images. They do this without any prior knowledge about Modern Baybayin Script/s.
- **Filipino text.** It can be Filipino words, phrases or sentence.

- **Image.** A Modern Baybayin script captured image in the format of .jpeg.
- **Image Processing.** Is one of the method the researcher will use in this study. It is also a method to convert an image into digital form and perform some operations on it, in order to get an enhanced image or to extract some useful information from it. It is a type of signal dispensation in which input is image, like video frame or photograph and output may be image or characteristics associated with that image.
- **Lingua Franca.**Also known as a **bridge language**, **common language**, **trade language**, **auxiliary language, vehicular language**, or **link language** is a language or dialects systematically used to make communication possible between people who do not share anative languageor dialect, particularly when it is a third language that is distinct from both native languages.
- **Machine Translator.**A translator of one language into another language by carried out of computer used.
- **Modern Baybayin.** Will be the output if the input is Filipino text and will be the input if the user wants to translate Modern Baybayin into Filipino text, also known as Badlit, is an ancient script used primarily by the Filipino people. Modern Baybayin is an indigenous Indic script that has been widely used in traditional Filipino domains. It is one of the many suyat scripts in the Philippines. The modern characters compose of 28 base characters or 128 all in all.
- **Natural Language Processing.**Natural language processing (NLP) is a branch ofartificial intelligencethat helps computers understand, interpret and manipulate human language. NLP draws from many disciplines, including computer science and computational linguistics, in its pursuit to fill the gap

between human communication and computer understanding. Since one of the function of this research is to translate text to image, the inputted text will understand and interpret by the system to translate into its equivalent output.

- ***N*-gram.** The algorithm gonna be use for matching syllabic rules for translation.
- **Script.** Particularly an alphabet or language that written in undefined symbol of handwriting.
- **Syllable.**A word that is being subdivided by the way of its proper spoken and pronunciation.
- **Windowed based.** Use of technology by the use of desktop computer, laptop, netbook or personal computer platform

Chapter 2

REVIEW OF LITERATURE AND STUDIES

2.1 Related Literature

2.1.1 Baybayin as National Writing of the Philippines

Recently, a bill seeking to declare Baybayin, a pre-Spanish writing system used in the early Philippines was signed by the House Committee on Basic Education and Culture. Known as the “National Writing System Act”, House Bill no. 1022 seeks to declare and promote Baybayin as the country’s national writing system, therefore promoting higher awareness on the legacy of Baybayin and as well as giving greater appreciation for it. Backed by the support of the advocacy group Baybayin, Buhayin, National Commission for Culture and the Arts (NCAA) and the Department of Education (DepEd) House Bill no. 1022 came into existence.

Once approved, signed and passed into law by the Senate, House Bill no. 1022 would mandate all Local Government Units (LGUs) in the country to use Baybayin in all their signage for street names, public buildings and facilities and other government establishments such as police and fire stations, community centers, hospitals and government halls. Once approved, House Bill no. 1022 shall also require all local manufacturing companies producing processed food products to inscribe Baybayin translation of their product into all of its label containers. Likewise, newspaper and magazine publishers would also be obliged to include a Baybayin translation of their name in their published works. (Guinto, 2018)

2.1.2 Old Baybayin Origin

Baybayin, in its time was primarily used to write diaries and poetries, and other personal writings. Shrouded by the lack of knowledge about the degree of extensiveness of Baybayin in the early time, Historians debate about the extent of its use or even the extent of its popularity and literacy among the Filipino natives during the pre-Hispanic era. William Henry Scott, a historian, cited evidences that Datus from the 1590s doesn't have the ability and literacy of signing affidavits, and evidences showing witnesses could not sign oaths in the 1620s leading to his argument that Filipino natives in the pre-Hispanic era are illiterate and has no literacy in writing.

Furthermore, Antonio Pigafetta, Ferdinand Magellan's chronicler in 1521made an article and wrote that Visayans were indeed literate. However, in contrary to Antonio Pigafetta's claim, the first governor of the Philippines, Miguel Lopez de Legazpi, pointed out that early native Filipinos are not illiterate. In his writings, he made it evident that the Visayanshas a system of literacy, a way of writing letters and characters, a way of writing which they learned from the Malays.

Century later, Legaspi's claim was supported by Jesuit historian Francisco Ignacio Alcina, who wrote that Visayanshadindeed a system of writing that they learned from the Tagalogs, who, in turn, learned it from the Borneans. Alcina called this system of writing the "moro writing".

Scholars namely, David Diringer, Fletcher Garner, and Isaac Taylor brought up theories about the origin of Baybayin. According to them, Asian ancient writing systems such as the Kawi Script from Java, Pallava Script from Sumatra,

Lontara Script from Sulawesi, Assamese Script from Bengal, and Cham Script from Champa may have a contribution from the origin of Baybayin.

Another possible theory came up that Baybayin writing system could possibly come from Sulus or Tausugs because in 1500s, Sulu was at the peak of its civi-cultural as well as politico-economic power before the appearance of Spaniards. In 1914, a Dean from University of Manila namely, Paul Rodriguez Versoza suggested to change the Baybayin into Alibata. Dean Paul Rodriguez Versoza coined Alibata from the first three letters of Arabic characters or alphabet; Alif,Ba, Ta, which was broadly known in Sulu Archipelago and Mindanao in ancien times. Dean Paul Rodriguez Versoza explained everything in his book entitled Pangbansang Titik nang Pilipinas, that was published way back 1939.

In early 1500s, the evangelization of natives by the Spanish priests depend heavily on the Baybayin. The book *Doctrina Cristiana* in Ilocano was introduced by Father Francisco Lopez.In 1621, the book which was an early record of Roman Catholic catechism published using the Tagalog form of Baybayin. However, Baybayin was slowly abandoned by people until it finally disappeared from use.

Samuel K. Tan a Tausug historian wrote that one of the first written literatures of Sulus or Tausugs was called *Luntar*, or Sulu annals. The *Moro Problem,* recollect being being given access to a library of ancient books by the Sultan of Sulu by Dr. Najeeb M. Saleeby. Most of the the books were religious manuscripts and books on law and magic. “There was no book on religion, law, or history in the possession of Datu Ali that I could not get, and the Sultan of Sulu placed his precious Luntarin my hands,” wrote Dr. Saleeby. (Jolo, 2014)

2.1.3 Modern Baybayin Origin

The addition of the set of alternative glyphs from their modern set and proposed reform was the main and immediately noticeable difference. The modern RA and the kawi based JA added, Mangyan pamudpod, x-kudlit, anusvāra, visarga, e/o kudlits, pallawa&padalaw-a marks, and other syllable-doubling marks; assigning them to the appropriate keys. What's not immediately noticeable until you use the font is the different aesthetic feel of the typeface. Their many tweaks that resulted on character element's uniformity (while keeping with the basic shapes of the old typeface) gives this font a definite modern feel, a lively touch, and youthful appeal. (Santos, 2016)

2.1.4 Old Baybayin Renovation into a Modern Baybayin

There are many people who tried to change the original Baybayin for Modern use. In 1620, one of the attempt to “improve” the Baybayin was rise when Fr. Francisco Lopez was about to publish the Ilokano Doctrina. He invented a new kudlit in the shape of cross, seeing the “limitations” of Baybayin. When the cross is placed under the script, the script would cancel the vowel after the consonant indicated on it.

Figure 7. **Baybayincharacter “B” with a kudlit**

Figure 2.1 shows the “B” script in Baybayin with kudlit. This simply shows that if there is no kudlit under it, it would be “Ba”. If you look at Baybayin as purely a writing system, this mutation is a natural process, the Spanish kudlit has made a comeback due people wanting a “complete” translation.

Figure 8. **Baybayin with a Spanish + kudlit**

Figure 2.2 shows the Baybayin with a Spanish + kudlit. Using the Spanish kudlit, it would look like this with the S:

Figure 9. **Baybayin with X and backslash**

Besides the Spanish + kudlit, An X and backslash after the glyph used as illustrated in Figure 2.3.

One of the many additions to the script championed by Bayani Mendoza de Leon was the backslash kudlit. De Leon released a manual full of instructions and history about Baybayin. While it does cover basic Baybayin, a lot of it is his proposal of a modern script with new characters.

Figure 10. **Baybayin represent Be, Bi, Bo, Bu script**

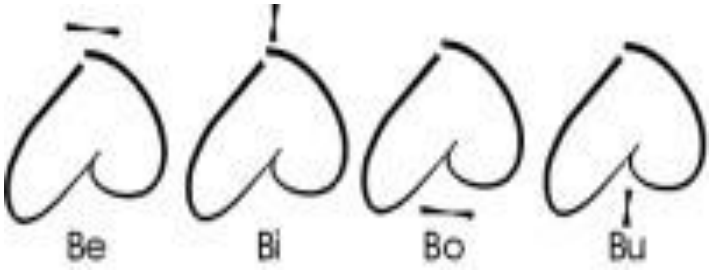


Figure 2.4 Show the Baybayin script of “Be”, “Bi”, “Bo” and “Bu”. As illustrated the backslash kudlit plays important role differentiating scripts in Baybayin. In addition to the backslash kudlit, the manual also goes into great detail

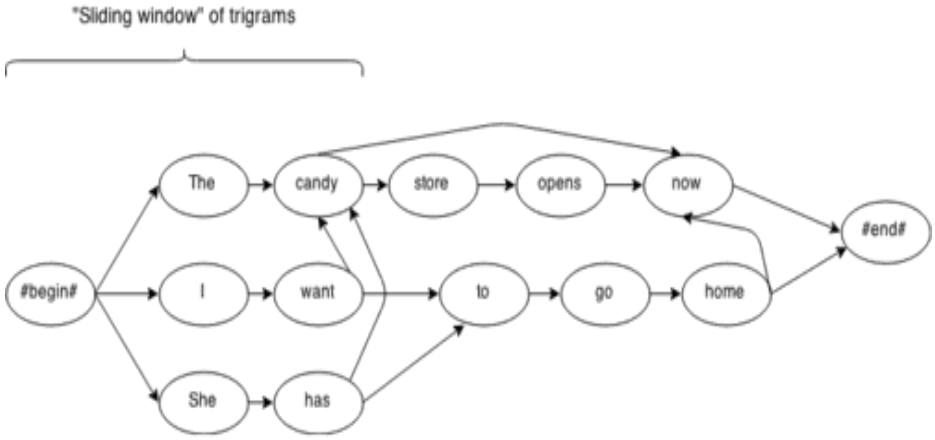
on the O vs U & E vs I issue. This looks to resolve some of the guesswork that's needed in order to read Baybayin.

Also covered are consecutive similar vowels and consonants, silent consonants and added consonant letters. (Kabuay, 2013)

2.1.5 N-gram

N-gram models are probabilistic models that assign probabilities on the "next" word in a sequence, given the n-1 previous words. The power of n-gram models was limited in natural language processing due to the fact that they cannot model deep dependencies, that is, the nth word only depends on the n-1 previous words; but they perform well on simple statistical demonstrations. (Math Kiser, 2015).

Figure 11. **N-gram visualization**



An n-gram can be best thought of as a moving window on a word. The n stands for a length. For exampe to n-gram the word quick, the results would depend on the length: (Elasticsearch, 2018)

Length 1 (unigram): [q, u, i, c, k]

Length 2 (bigram): [qu, ui, ic, ck]

Length 3 (trigram): [qui, uic, ick]

Length 4 (four-gram): [quic, uick]

Length 5 (five-gram): [quick]

Using a technique called N-gram analysis, it is possibly to very quickly extract sequences of characters from text and calculate the frequency of occurrences with the text. (ThreatLabz, 2010)

The application of n gram can be use in given the probabilities of a sentence it can determine the likelihood of an automated machine translation being correct, it could predict the next most likely word to occur in a sentence, it could automatically generate text from speech, automate spelling correction, or determine the relative sentiment of a piece of text. (Kevin Sookocheff, 2015)

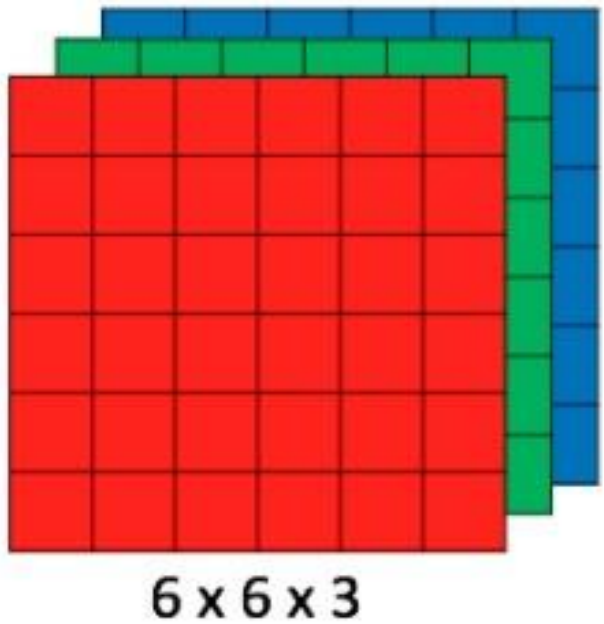
2.1.6 Convolutional Neural Networks

In neural networks, Convolutional Neural Network (ConvNet or CNN) is one of the main categories for image recognitions and images classifications and areas such as objects detections and face recognition are some of the fields where CNNs are widely used.

CNN image classifications works by taking an input image, processes it and classifies it under certain categories (Eg., Dog, Cat, Tiger, Lion). Computers sees an input image as array of pixels and it depends on the image resolution. Based on the image resolution, it will see h x w x d(h = Height, w = Width, d = Dimension).

Eg., An image of 6 x 6 x 3 array of matrix of RGB (Red, Green, Blue) (3 refers to RGB values) and an image of 4 x 4 x 1 array of matrix of grayscale image.

Figure 12. **Array of RGB (Red, Green, Blue) Matrix**



Technically, deep learning CNN models to train and test, each input image will pass it through a series of convolution layers with filters (Kernals), Pooling, fully connected layers (FC) and apply Softmax function to classify an object with probabilistic values between 0 and 1. The below figure is a complete flow of CNN to process an input image and classifies the objects based on values(Prabhu, 2018).

Convolutional Layers. Convolution is the first layer to extract features from an input image. Convolution preserves the relationship between pixels by learning image features using small squares of input data. It is a mathematical operation that takes two inputs such as image matrix and a filter or kernel(Prabhu, 2018).

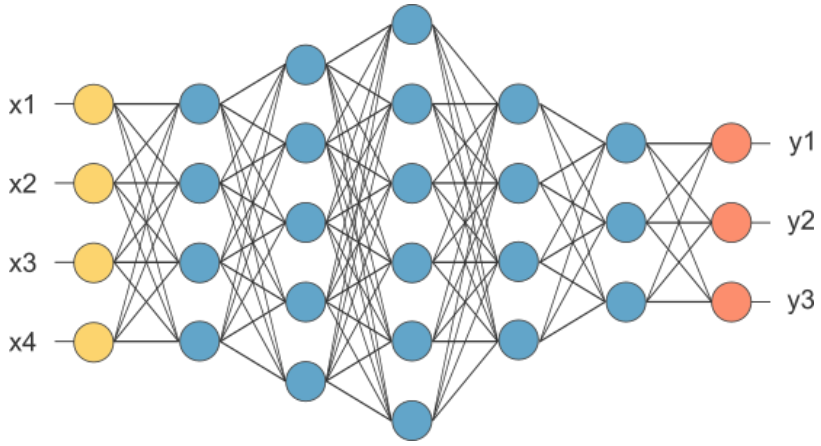
Pooling Layers. Pooling layers section would reduce the number of parameters when the images are too large. Spatial pooling also called subsampling or downsampling which reduces the dimensionality of each map but retains the important information. Spatial pooling can be of different types:

- Max Pooling
- Average Pooling
- Sum Pooling

Max pooling takes the largest element from the rectified feature map. Taking the largest element could also take the average pooling. Sum of all elements in the feature map call as sum pooling (Prabhu, 2018).

Fully Connected Layers.

Figure 13. **Neural Network with Convolutional Neural Network**



The layer we call as FC layer, Prabhu flattened their matrix into vector and feed it into a fully connected layer like neural network (Prabhu, 2018).

2.1.7 Syllabication

The division and partition of a word into syllables, in written or spoken form is called syllabication or syllabification. For most countries, the way that the syllables are uttered is also the basis of its syllabication once turned into its written form. But, potentially due to the weak connection between the letters in the spelling and the way that it is said, written syllabification is mostly based on morphological or etymological principles instead of phonetic concept (Abed, 2012).

Fern, a reading specialist in an elementary school often have students that struggle with decoding, or the area of reading involves sounding out words on a page. According to Fern there four important and most common rules to remember when decoding: (1) In a word, when a vowel is in between two consonants, the syllables should be segmented between the consonants. A good example of this rule include un/der, san/dy, or pig/let. (2) When two or more consonants is present in a word, the syllables should be divided so that the blends stay intact. Examples application of this second rule includes en/trance, mon/ster, or sub/tract. (3) In a word when two vowels have a consonant in between, the word should be divided after the first vowel, before the subsequent consonant. Some examples are mi/nus, a/ni/mal, and cu/rate. (4) If a word contains a double consonant, the syllables should be divided between the two consonants. Examples include hap/py, lit/tle, and diz/zy. (Fern, 2013).

In speaking the Filipino language, words are **syllabicated or broken down by syllable** in a slightly different way than in English. The process of how

Table 1

Different forms of Filipino syllables and examples

Syllable Form	Examples (in bold)
P	ako (**a**·ko), edad (**e**·dad), i·sa (**i**·sa), noo (no·**o**), oso (**o**·so), ubo (**u**·bo)
KP	ba·ka (**ba**·**ka**), dugo (**du**·**go**), relo (**re**·**lo**), si·la (**si**·**la**)
PK	akda (**ak**·da), isda (**is**·da), ugnay (**ug**·nay), uod (u·**od**)
KPK	aklat (ak**lat**), bundok (**bun**·**dok**), dahon (da·**hon**), isip (i·**sip**)
KKP	bloke (**blo**·ke), braso (**bra**·so), droga (**dro**·ga), gripo (**gri**·po), plato (**pla**·to)
PKK	abstrak (**abs**·trak), ekstra (**eks**·tra)
KPKK	**disk**, **fern**, gibmefayb (gib·me·f**ayb**)[1], iskeyt (is·**keyt**), isport (is·**port**), **keyk**, kontest (kon·**test**), **nars**, **test**
KKPK	blangko (**blang**·ko)[2], brilyo (**bril**·yo), **dyip**, dyakpat (**dyak**·pat)[3], ekspres (eks·**pres**), ispred (is·**pred**), isprey (is·**prey**), **kyut**, plastik (**plas**·tik), playwud (**play**·wud), prinsesa (**prin**·se·sa), **swak**, **trak**, **tren**, **tsek**, **tres**, trangkaso (**trang**·ka·so)[2], tsamba (**tsam**·ba), tsismis (**tsis**·mis)
KKPKK	**bleyd**, **breyk**, **breyslet** (**breys**·let), **treynta** (**treyn**·ta), **tsart**
KKPKKK	**shorts**

2.2 Related Studies

2.2.1 Translation of Writing System into a form of text

In order to easily understand and teach other form of languages many researchers came up to an idea of translating words or writing systems. As statedin the thesis paper from the School of Engineering Technology West Bengal University of Technology, India, entitled, An Algorithm for Japanese Character Recognition, theyproposed a geometry-topology based algorithm for Japanese Hiragana character recognition. The values of quantities both for the target and the template images are computed and math is made with the character having the

minimum Euclidean distance with the average accuracy of 94.1%. (Das et al., 2015).

Another research focused on Thai Characters, according to the thesis paper from Stanford University, USA, entitled Recognition of Thai Characters and Text from Document Templates they created an image processing system that takes in an input image of a Thai text document, since it is an image processing kind of system it needs a phone camera. Taken from a phone camera and outputs a matching between each Thai character detected in the document to a particular Thai template character. The output could then be used to input into translation software to translate the Thai sentences out automatically without the need of a human translator. (Asavareingchai et al., 2016). Using XOR + Resize + WDS algorithm is the most favorable result with the 91% for clean (same font) 80% for clean (different font), 95% for test (same font), 83% for test (different font).

On another research that focuses on hieroglyphics writing from the University of Amsterdam, Netherlands, entitled Automatic Egyptian Hieroglyph Recognition by Retrieving Images as Texts, they proposed an approach for automatically recognizing ancient Egyptian hieroglyph from photographs and segmented a large collection of nearly 4,000 hieroglyphs. They determine the reading order and subsequently evaluate 5 visual descriptors in 3 different matching schemes to evaluate visual hieroglyph recognition and with correctly annotated of hieroglyphics of 85.5 % automatic detection according to Pascal VOC overlap criteria. (Franken et. al, 2013).

On a similar research that proposed an image based document translation using an error correction model to correct those words who are recognized incorrectly from OCR output. The research is according to the writings from

Computer Science Camegie Mellon University, USA, entitled Enhancing Imagebased Arabic Document Translation Using a Noisy Correction Model. They train their correction model from synthetic data with different fonts and sizes to simulate real world situations and further enhance their recognition with bigrams to improve their word segmentation error correction. For instance, in an experiment using Arabic Transparent Font, the BLEU score increases from 18.70 to 33.47 with the use of their noisy channel model (Chang et al., 2010).

On a different approach, there are also researchers that studied the stenography writing translation into text, according to a thesis paper from the Polytechnic University of the Philippines, Philippines, entitled Gregg Stenography Image-To-Text Conversion and Vice Versa, they presented a tool to recognize written Gregg Stenography from an image to be translated into English text using Artificial Neural Networks (ANN) for pattern recognition and Natural Language Processing (NLP) algorithms for conversion of English text into its corresponding Gregg Shorthand. The research concerns in finding the completeness and correctness of the system's output, in text-to-image and vice versa using the specificity and sensitivity as the statistical treatment. With a degree of correctness of 10.72778 and 11.06389 for the degree of completeness (Casimiro, et al. 2013).

In accordance to the paper by Obaid et. al. entitled Handwritten Text Recognition System Based on Neural Network, handwritten text recognition is still an open research issue in the domain of Optical Character Recognition (OCR). Their paper proposes an efficient approach towards the development of handwritten text recognition systems. 3-layer Artificial Neural Network (ANN) is utilized in their Paper using supervised learning approach. The choice of optimal feature vectors greatly affect the accuracy of any text recognition system therefore

bit map representation of input samples are utilized as feature vector. The feature vectors are first pre-processed and then applied to the ANN along with the generated target vectors; that are generated on the basis on input samples. 55 samples of each English alphabet are used as a ANN training process in order to make sure the general applicability of system towards new inputs. Two different learning algorithms are utilized in this paper. Additive image processing algorithms are also developed in order to deal with the multiple characters input in a single image, tilt image and rotated image. The trained system provides an average accuracy of more than 95 % with the unseen test image. (Obaid et. al., 2016)

According to the paper by Prof. Dr. Klaus Schneider entitled Optical Character Recognition - A Combined ANN/HMM Approach. They presented a new OCR method for degraded text recognition by introducing a combined ANN/HMM OCR approach. The approach provides significantly better performance in comparison with state-of-the-art HMM based OCR methods and existing open source OCR systems. In addition, the thesis introduces novel applications of ANNs and HMMs for document image preprocessing and recognition of low resolution text. Furthermore, the thesis provides psychophysical experiments to determine the effect of letter permutation in visual word recognition of Latin and Cursive script languages. Mostly, previous attempts in combining the HMMs and ANNs were focused on applying ANNs as approximation of the probability density function or as a neural vector quantizer for HMMs. These methods either require combined NN/HMM training criteria [ECBGMZM11] or they use complex neural network architecture like time delay or space displacement neural networks [BLNB95]. The presented system achieves a 30% reduction in error rate as compared to Google's Tesseract OCR system [Smi13] and 43% reduction in error as compared to

OCRopus OCR system [Bre08], which are the best open source OCR systems available today. The proposed method resulted in more than 95% script recognition accuracy on various multi-script documents at connected component level and 100%-page orientation detection accuracy for Urdu documents (Schneider, 2014).

According to the paper from ICCC International Digital Design Invitation Exhibition presented by Recario, Reginald Neil C. et al. entitled, An Automated Reader Philippine Baybayin Scripting Image Processing Method. The description of their work is that there are more recent discoveries in the Philippine artifacts bearing the Baybayin scripture and with the institution of the National Script Act of 2011, they presented an automated reader for the Baybayin scripts. The system is meant to give significant impact in promoting and reviving the use and relevance of the Baybayin script. The system generally analyses an input image of a Baybayin script and outputs the equivalent syllables of the script. The system is independently tested using the Berthod and Maroy's Primitives and Freeman Chain Coding. They used independently implemented by means of OpenCv (Recario et. al., 2011).

2.2.2 Translation of Image to Text using Convolutional Neural Network

As stated in the paper by Zohra Saidane entitled Image and video text recognition using convolutional neural networks, there are four steps for the recognition of the text: (1) detecting the presence of the text, (2) localizing of the text, (3) extracting and enhancing the text area, and finally (4) recognizing the

content of the text.Their work focus on this last step and we assume that the text box has been detected, located and retrieved correctly. This recognition module can also be divided into several sub-modules such as a binarization module, a text segmentation module, a character recognition module. They focused on a particular machine learning algorithm called convolutional neural networks (CNNs). These are networks of neurons whose topology is similar to the mammalian visual cortex. CNNs were initially used for recognition of handwritten digits. They were then applied successfully on many problems of pattern recognition. They proposed in this thesis a new method of binarization of text images, a new method for segmentation of text images, the study of a convolutional neural network for character recognition in images, a discussion on the relevance of the binarization step in the recognition of text in images based on machine learning methods, and a new method of text recognition in images based on graph theory.For the alphabetic characters, the character recognizer has an 87% success rate while, for the alphanumeric characters, it drops down to 83% success. When all of the punctuation is included, the accuracy drops to 68%. When this system is combined with a language model using dynamic programming, the overall performance is in the vicinity of 80-95% word accuracy on pages captured with a 1024 × 768 webcam and 10-point text (Saidane, n.d.)

In accordance to the paper by Andew Wu from Department of Computer Science in Stanford University entitled End-to-end Text Recognition with Convolutional Neural Network, full end-to-end text recognition in natural images is a challenging problem that has recently received much attention in computer vision and machine learning. Traditional systems in this area have relied on elaborate models that incorporate carefully hand-engineered features or large amounts of

prior knowledge. In this thesis, I describe an alternative approach that combines the representational power of large, multilayer neural networks with recent developments in unsupervised feature learning. This particular approach enables us to train highly accurate text detection and character recognition modules. Because of the high degree of accuracy and robustness of these detection and recognition modules, it becomes possible to integrate them into a full end-to-end, lexicon-driven, scene text recognition system using only simple off-the-shelf techniques. In doing so, we demonstrate state-of-the art performance on standard benchmarks in both cropped-word recognition as well as full end-to-end text recognition. The accuracy of their study is 84%. Part of this gain in performance was due directly to the higher accuracy of their character classifier. Compared to the character classifier, which obtained an accuracy of 64% on the ICDAR 2003 character sets, their character classifier obtained an accuracy of almost 84% (Wu, 2012).

2.2.3 Translation of text using N-Gram

In accordance to the paper of Canvar and Trenkle entitled N-Gram-Based Text Categorization, described here is an N-gram-based approach to text categorization that is tolerant of textual errors. The system is small, fast and robust. This system worked very well for language classification, achieving in one test a 99.8% correct classification rate on Usenet newsgroup articles written in different languages. The system also worked reasonably well for classifying articles from a number of different computer-oriented newsgroups according to subject, achieving as high as an 80% correct classification rate. There are also several obvious directions for improving the system's classification performance in those cases where it did not do as well. The system is based on calculating and comparing

profiles of N-gram frequencies. First, we use the system to compute profiles on training set data that represent the various categories, e.g., language samples or newsgroup content samples. Then the system computes a profile for a particular document that is to be classified. Finally, the system computes a distance measure between the document's profile and each of the category profiles. The system selects the category whose profile has the smallest distance to the document's profile. The profiles involved are quite small, typically 10K bytes for a category training set, and less than 4K bytes for an individual document. Using N-gram frequency profiles provides a simple and reliable way to categorize documents in a wide range of classification tasks (Canvar et al., n.d.)

From the paper of conference on 2012 International Conference on Asian Language Processing, IEEE entitled An Approach to N-Gram Language Model Evaluation in Phrase-Based Statistical Machine Translation, N-gram language model plays an important role in statistical machine translation. Traditionally methods adopt perplexity to evaluate language machine translation. In their paper, they propose a novel method namely a bag-of-words decoding to evaluate n-gram language models in phrase-based statistical machine translation (Su et al., 2012).

From the thesis paper from DAV College, India, entitled N-Gram MachineTranslation System for English Punjabi Transliteration, He make a system that is made in such a way that it first learns from existing examples stored in the database and then with the help of N-Gram approach he translate the proper nouns of English language into its equivalent Gurumukhi Script. With a accuracy of 97% for the training of 15,000 and test of 2000. The system is tested on the various names belongs to various regions and overall accuracy of the system is very good. (Jindal, 2015)

From another research paper from Polytechnic University of the Philippines, Philippines, entitled Filipino Text BaybayinAlphasyllabary Translator, the researcher developed a mobile-based translator of Filipino Text to Baybayin Character for people to easily understand the alphasyllabary system. The study is for people who are interested in studying and teaching baybayin alphasyllabary system. The researchers used n-gram as the algorithm in obtaining the corresponding syllabication rule of a word. The proponents tested the accuracy of syllabication in translating words and sentences in terms of affixation, cluster and diphthongs with an accuracy of 99.2%. (Pedroza et al., 2016)

2.3 Synthesis of The Study

In the Thesis Paper of Raymark G. Pedroza and Mae Antoinette P. Simpao entitled "**A Mobile-Based Filipino Text to Baybayin Characters Translation"** from Polytechnic University of the Philippines, they proposed a mobile-based application/systemthat will translate Filipino words/s, phrases or sentences in the form of text into Old Baybayin characters/script that achieved 99.2% accuracy. The system architecture is consisted of input text/sentence, preprocessing, syllabication, character generation, and output. The input of text/sentence will be done by the user. The preprocessing isconsisting of tokenization and removal of symbols, in which tokenization is if the input is more than two words the system will segment it into word. The syllabication is the process of forming or dividing words into syllables and they use n-gram algorithm in obtaining the corresponding syllables rules of the text. The character generation is generating the baybayin character according to the syllable extracted from the previous process and they use decision tree in checking the appropriate baybayin character. Then the last process is showing the output or results. They use affixation (prefix, suffix,

circumfix, infix), cluster and diphthongs as their parameters for computing the accuracy of the proposed system. The baybayin characters/script they use is the old baybayin which consisted of 17 base characters

.

In this paper, for the presence of loving our ancient writing system, we are working on a window-based application/system written using Java programming language that will also convert Filipino text in the form of text into Modern Baybayin characters/script and vice versa. The system will use Modern Baybayin Script instead of Old Baybayin Script. The OldBaybayin script only consists of 17 base characters/script and 73 all in all while the Modern Baybayin consists of more than 28 base characters and 128 all in all made by the modification of Spaniards. The additional characters/script are the BaybayinViramas and Marks, Optional Characters, Additional Character for modern use. The parameter for computing the accuracy of the Tagalog words to Modern Baybayin Script will be the syllabicated word/s and for the Modern Baybayin Script to Tagalog Text the parameters would be the Modern Baybayinscript itself.

Table 2

Synthesis Matrix for Text to Image Translation

Study	**Approach**	**Accuracy Rate**	**Database Presence**	**Research Aim**

N-Gram-Based Text Categorization	Compute profiles on training set data that represent the various categories, e.g., language samples or newsgroup content samples. Then the system computes a profile for a particular document that is to be classified. The system selects the category whose profile has the smallest distance to the document's profile.	80%	Yes	Language Detection and Article Classification
N-Gram MachineTranslation System for English Punjabi Transliteration	The system first learn from existing examples stored in the database and then with the help of N-Gram approach he translate the proper nouns of English language into its equivalent Gurumukhi Script	97%	Yes	Translate the proper nouns of English language into its equivalent Gurumukhi Script
Filipino Text BaybayinAlphasyllabary Translator	Used n-gram as the algorithm in obtaining the corresponding syllabication rule of a word. Then use decision tree for identifying the right Baybayin script. Old Baybayin Script was used for the translation.	99.2%	No	Translate Filipino text into Baybayin Characters
Filipino Text to Modern Baybayin Script in Image Form Using N-Gram & Modern Baybayin Script in Image Form to Text Using Convolutional Neural Network	Implemented n-gram algorithm in obtaining the corresponding syllabication rule of the input. Utilized decision tree for identifying the right Modern Baybayin script. Used the Modern Baybayin Script for translation	-	Yes	Translate Filipino Text to Modern Baybayin Script using Ngram algorithm

Based on the table above, the N-Gram-Based Text Categorization achieved 80% accuracy rate, the N-Gram Machine Translation System for English

Punjabi Transliteration achieved 97% accuracy rate and the Filipino Text BaybayinAlphasyllabary Translator which was the closest study to relate achieved 99.2% accuracy rate. The existing previous research doesn't have the validation of words for the filtration of the input. The different studies in the table used n-gram as one of their algorithm, one for language detection and article classification, one for translating the proper nouns of English language into its equivalent Gurumukhi Script and the other for translating Filipino text into Baybayin Characters algorithm. From this, the researchers can conclude that they all achieved an acceptable high accuracy rate for their system and that n-gram was a good algorithm for translating one language to another language. For this reason, the researchers will use the ngram algorithm for translating Filipino text to Baybayin Script and will have the validation of words for filtering the input.

Table 3

Synthesis Matrix for Image to Text Translation

Study	Method	Accuracy	Research Aim
An Algorithm for Japanese Character Recognition	Size-TranslationRotation-Invariant Character Recognition and Feature vector Based - STRICRFB algorithm	94.1 %	Character recognition of Japanese Hiragana characters

Recognition of Thai Characters and Text from Document Templates	XOR + Resize + WDS algorithm	Average of 87.25 %	Matching of Thai text document and a Thai template character
Automatic Egyptian Hieroglyph Recognition by Retrieving Images as Texts	CNN	85.5 %	Automatically recognizing ancient Egyptian hieroglyph from photographs
Gregg Stenography Image-ToText Conversion and Vice Versa	ANN	10.72778 degree of correctness and 11.06389 degree of completeness	Recognize written Gregg Stenography from an image to be translated into English text
Handwritten Text Recognition System Based on Neural Network	ANN	95%	Proposes an efficient approach towards the development of handwritten text recognition systems
Optical Character Recognition - A Combined ANN/HMM Approach	ANN/HMM OCR	95%	Present a new OCR method for degraded text recognition by introducing a combined ANN/HMM OCR approach
An Automated Reader Philippine Baybayin Scripting Image Processing Method	Berthod and Maro's Primitives and Freeman Chain Coding		Analyses an input image of a Baybayin script and outputs the equivalent syllables of the script.
Filipino Text to Modern Baybayin Script in Image Form	CNN	-	Translation of modern ModernBaybayin script
Using N-Gram & Modern Baybayin Script in Image Form to Text Using Convolutional Neural Network			obtained from an image into its equivalent filipinoText.

The table above shows that the application of image processing for character recognition for translation is fairly common. As mentioned above, previous researchers used different approach for the translation of image data into text such as STRICR-FB algorithm, XOR + Resize + WDS algorithm, CNN, ANN and HMM which all achieved a fairly high accuracy in correctly recognizing characters.

Because of the popularity of ANN in this area, the researchers decided to try and use a different approach, it is to utilize and exploit the advantages of another type of deep neural network, namely CNN or Convolutional Neural Network for this research in hopes of finding that CNN would perform better than the expected output if ANN was used and potentially proving that CNN is the better type of neural network suited for optical character recognition and specifically for the study's uncommon type of character recognition.

Chapter 3

METHODOLOGY

This Chapter focuses on tackling the specific procedures or techniques used to identify, select, process, and analyze information about the Research. The Research Design, System Architecture, Sources of Data and its procedure and Statistical Treatment of Data were stated here.

3.1 Research Design

Experimental research is a traditional type of research that is quantitative in nature whose initial approach or move is to scrutinize causal (cause/effect) relationships and to examine the connection of one variable to another. It is also that type of research conducted with scientific approach, where a set of variables are kept constant while the other set of variables are being examined as the subject of experiment. Briefly, experimental research is used to test or compare set of variables needed in a certain research. (Bhat, 2018)

In this study, the constant variables would be the system input such as the scanned handwritten images and the Filipino text input for it will not undergo any treatment therefore remaining unchanged throughout the entire system process while the translated system outputs would be the subject of experiment of this research for it will serve as the basis of comparison and computation of the accuracy and reliability of the system.

3.2 Sources of Data

The researchers conducted an interview to an expert named, Mr. NiloBoquirin a Teacher I from Panghulo National High School for them to have an insight about Modern Baybayin Characters/Script and also to gain knowledge that the researches will use to

build this study. The knowledge or data gathered by the researchers used to build this research. The researchers prepared list of words to be examine by their system using the list of Filipino Text and handwritten images of Modern Baybayin script. They prepared list of Filipino text to be stored in database.

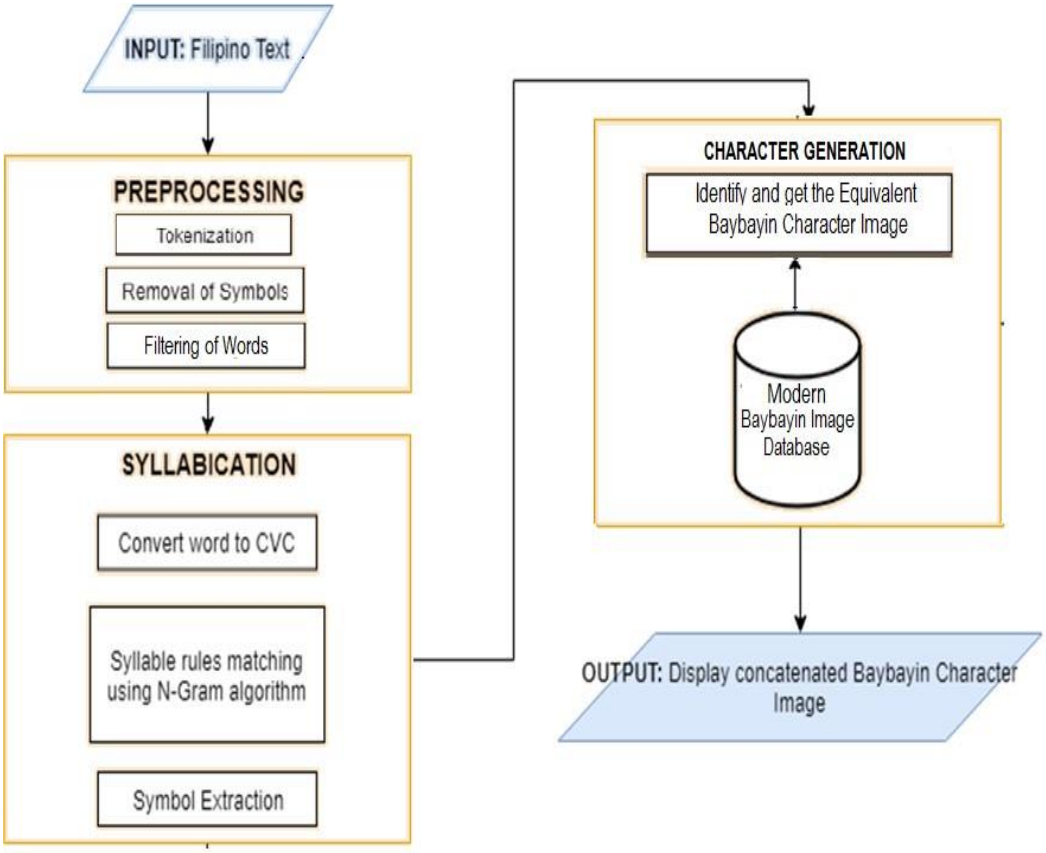
Proportional and non-proportional quota sampling was used by the researchers to select data. Proportional and non-proportional quota sampling, members are sampled until exact proportions of certain types of data are obtained or until sufficient data in different categories is collected (CIRT, n.d.). Using Proportional and nonproportional quota sampling, the researchers prepared data for text to image translation and vice versa.

3.3 Research Instrumentation

3.3.1 Software/Hardware Tools

3.3.2 System Architecture

Figure 14. **System Architecture of Text to Image Translation**



A. Input Filipino Text – the phase where the proponents input Filipino text.

B. Preprocessing – this process consists of tokenization which will segment words, sentences or phrases into set of one word and removal of other invalid symbol (‘@’, ‘_‘, ‘|‘, ‘#’, ‘$’, ‘%’, ‘^’, ‘&’, ‘=’, ‘+’, ‘*’, ‘<’, ‘>’, ‘`’, ‘~’, ‘[‘, ‘]’, ‘(‘, ‘)’, ‘{‘, ‘}’, ‘:’, ‘;’, “‘”, “‘”, ‘\’ ,’/’) and invalid words.

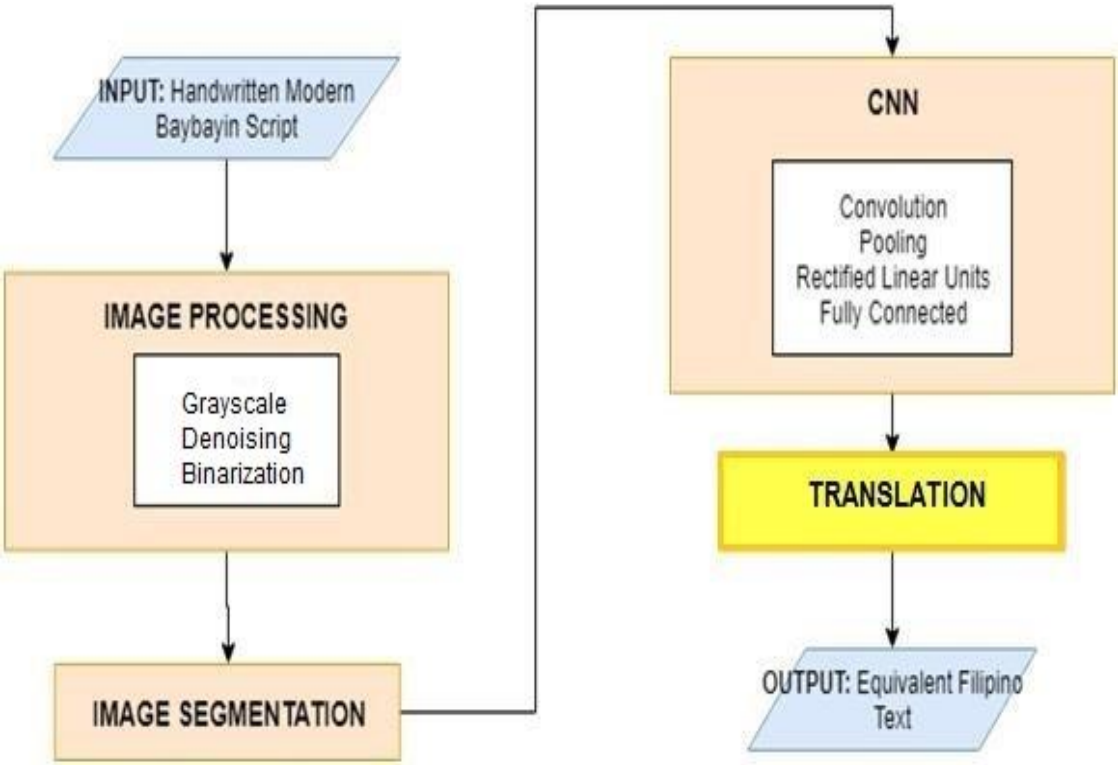
C. Syllabication – this process consists of converting the set of one word into

CVC, wherein each letter will be converted into C if it's Consonant or V if it's Vowel. After converting, the N-gram algorithm will be used for syllable rules matching. The output of this process is the syllabicated word/s.

D. Character Generation – To select and identify the equivalent Modern Baybayin Character of each syllables and get the Modern Baybayin Character Image from the database.

E. Output – this will show the concatenated equivalent Modern Baybayin Character Image of the syllables that will form the word/s.

Figure 15. **System Architecture of Image to Text Translation**



A. Handwritten Modern BaybayinScript Image – This is the first phase of Modern Baybayin to text translation. The user or proponents will prepare a hand written Modern Baybayin Character/s written in a white clean bond paper using black pen.

B. Image Preprocessing – the scanned handwritten characters will now undergo preprocessing to improve the image data. The researchers used four different techniques for the preprocessing of the images. These four preprocessing techniques are the following:

- Grayscale - The process of converting the image into black-and-white in order to detect edges more accurately. For this are the proponents will be using Luminosity Method.
- Binarization – the process of converting the grayscale image into binary image. To address this process the proponents will be using Gatos Method.
- Denoising – the process of removing unnecessary noises from the image. For this part of the pre processing the proponents will be using Median Filter.

C. Image Segmentation – process of partitioning or dividing the image into multiple segments. For this process the proponents will be utilizing Horizontal Projection method.

D. CNN – after the features of the image have been extracted it will be fed to the Convolutional Neural Network for classification before outputting the final result. Generally there were 4 layers inside the CNN (Convolutional Neural Network), and these were the following:

- Convolution – extract features from the input image.
- Pooling – reduces the dimensionality of each feature map.
- Rectified Linear Units (ReLU) - replaces all negative pixel values in the feature map by zero.

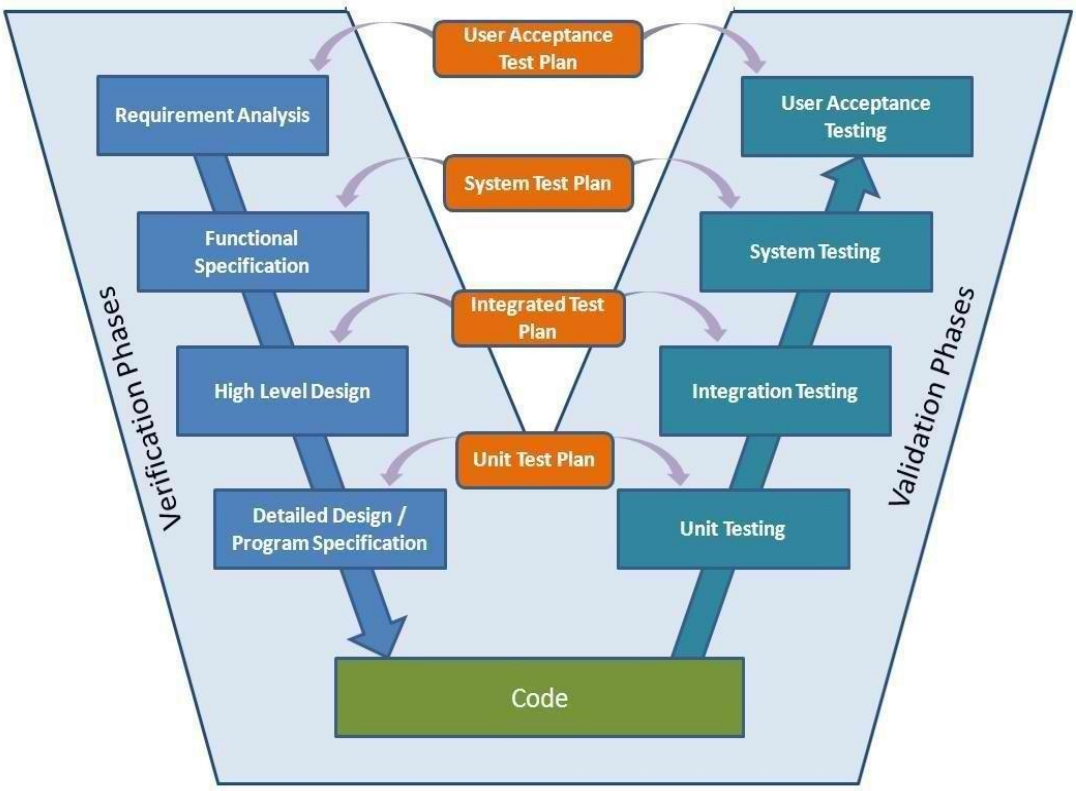
- Fully Connected Layers.-a traditional Multi Layer Perceptron that uses the features gathered for classifying the input image into various classes based on the training dataset.

E. Translation – after the image has been classified, it will now be translated according to its corresponding text.

F. Equivalent Filipino Text – the final output of the system from the handwritten modern baybayin script.

3.4 Development Details

Figure 16. **V Model**



Also known as verification and validation model, V modelis a type of SDLC model where process execution happens in a V-shaped sequential manner. V model is based on the association of a testing phase for each corresponding development stage. Like the waterfall model, phases in a V model must be completed first before starting the next phase of the development.

- **Requirements Analysis| Acceptance Test**

  Requirements Analysis - all possible requirements needed for the development of the system is gathered, defined and analyzed.

  Acceptance Test - testing of the product in the user environment to check and verify whether the created system meets its proposedrequirements and now ready for propose system deployment.

- **Functional Specifications | System Testing**

  Functional Specifications- designing of the entire system. Techniques and different approaches to be used for the entire system were finalized.

  System Testing- testing of the entire system and mainly focused on system output and performance.

- **High-level Design | Integration Testing**

  High-level Design - focused on the architectural design of the system. Subprocesses not included in functional specifications were finalized in this phase.

  Integration Testing-combining and testing of the two core processes of the system: the conversion of Filipino words to Modern Baybayin and the conversion of Modern Baybayin to Filipino text.

- **Detailed Design| Unit Testing**

Detailed Design - focused on the design of the smallest details of the system including its units and functional logics.

Unit Testing – testing of the smallest units of the system to early detect and eliminate bugs

- **Code/Coding Phase** - phase for the development of the entire system based on the designs that they made during the previous phases.

3.5 Research Instrument

Software

- **Open CV –** an open source library commonly used for computer vision, image processing and machine learning
- **Java &Netbeans IDE –** the researchers will use Java programming language together with Netbeans IDE for the development of the entire system including the training of the Convolutional Neural Network and the system interface itself.
- **Neural Network Tool –** provides algorithms, pre-trained models, and apps to create, train, visualize, and simulate both shallow and deep neural networks.
- **Matlab -** a high-performance language for technical computing. It integrates computation, visualization, and programming in an easy-to-use environment where problems and solutions were expressed in familiar mathematical notation.
- **MySQL Workbench** –used for the image of modern baybayin repository of data.

Hardware Tools

- **Experimental paper –** used in conducting results during testing of accuracy and reliability made by the researchers.

- **Digital Computer –** the platform where the system needs to be installed in order to test it and will be used for the development of the system.
- **Pen & White clean paper –** used for manually writing the Modern Baybayin Characters to be translated into text.
- **Image Scanner -** a device used to optically scan images that contains hand written Modern Baybayin Characters/Script and converts it to a digital image.

3.6 Data Generation/Gathering Procedure

- The researchers prepared list of Filipino text to be stored in training data database.
- The researchers prepared list of Filipino text and handwritten image of Modern Baybayin Script to be examine by their system.
- The researchers determined the methods, procedures, and techniques that will produce the expected output of the study.

3.7 Statistical Treatment of Data

The objective of this research was to obtain the right accuracy and reliability in translating Filipino text to Modern Baybayin Characters/Script and vice versa.

Accuracy

To calculate the system's accuracy, the data gathered that was examine using the system used.

$$\mathbf{ER} = ((\text{total} - \text{RC}) / \text{total})$$

$$\mathbf{RAP} = (1 - \text{ER}) * 100\%$$

Where:

ER = Error Rate

RAP = Recognition Accuracy Percentage

RC = Recognized Correctly

Total = Total number of Input

Reliability

The formula to get the Reliability of this Research, the researchers use the Pearson R test.

$$\mathrm{r} = \frac{\mathrm{N}\Sigma\mathrm{xy} - (\sum\mathrm{x})(\Sigma\mathrm{y})}{\sqrt{[\mathrm{N}\Sigma x^2 - (\Sigma x)^2][N\Sigma y^2 - (\Sigma y)^2]}}$$

N = number of paired scores

Σxy = sum of the product of paired scores

Σx = sum of the product of x scores

Σy = sum of the product of y scores

Σx^2 = sum of squared x scores

y^2 = sum of product y scores

Chapter 4

RESULTS AND DISCUSSION

This chapter, the researchers present the result of the implementation of translation 100 Filipino text into Modern Baybayin script image and 100 Modern Baybayin Image to be translated into Filipino text using the developed system. The formula stated in the Chapter 3 were be used to evaluate and present the result of testing. It will show the analysis and discussion of the accumulated information through the implementation process.

This chapter also discusses the table results and interpretation of the problem for solving the accuracy in terms of:

1. What is the degree of accuracy of translation with respect to:

    a. Filipino Text to Modern Baybayin script?
    b. Filipino Text Syllable to Modern Baybayin Script?
    c. Modern Baybayin Script to Filipino Text Syllable?
    d. Modern Baybayin script to Filipino text?

Table 4

Result of the translation of Filipino Text to Modern Baybayin Script

Number of input	Total Correct Filipino Text	Accuracy
100	95	95%

Table 4.1.1 Shows the result of the translation of Filipino Text Syllable to Modern Baybayin Script. The researchers prepared 100 inputs that were validated and modified by their expert to test the accuracy of the developed system. The table above shows correct translated Filipino Text translated to Modern Baybayin Script. Thus, resulting to an accuracy of 95% overall. The researchers computed those accuracy using the Accuracy rate formula and Error rate formula. See Appendix I in Appendix A, Table A to see a table that showed detailed input data and how it tallied.

Table 5

Result Record of Filipino Text Syllable to Modern Baybayin Script Translation

Number of input Syllable	Total Correct Filipino Text	Accuracy
4,250	4,313	98.6%

The researchers prepared 100 inputs that consist of simple sentence, paragraph and etc. See Appendix A to see a table that showeddetailed input data and how it tallied. Overall, it consists4,313 syllable in total. The table above shows

total score syllable and total number of input correctly translated by the system and it is resulted to 4,250 over 4,313 syllables which resulted to 98.6% accuracy. The researchers computed those accuracy using the Accuracy rate formula and Error rate formula.

Table 6

Result Record of Modern Baybayin Script to Filipino Text Translation

Number of input	**Total Correct Baybayin Script**	**Accuracy**
100	72	72%

Table 4.1.2 shows the Result Record of Modern Baybayin Script to Filipino Text in terms of accuracy per syllable and per text. The researchers prepared 100 inputs that were validated and modified by their expert to test the accuracy of the developed system. The table above shows correct translation of Modern Baybayin Script to Filipino Text. There were 72 correclty translated Modern Baybayin Script. The developed system recognized whatever writings in the input script whether it is a non-sense or just a drawing but it will output a wrong output since it has no equivalent characters. Thus, resulting to an accuracy of 72% overall. The researchers computed those accuracy using the Accuracy rate formula and Error rate formula.

Table 7

Result Record of Modern Baybayin Script Syllable to Filipino Text Translation

Number of input Syllable	Total Correct BaybayinScript Syllable	Accuracy
2,244	2,053	91%

The researchers prepared 100 inputs that consist of 2,244 syllable in total. The table above shows total score per syllable that is being correctly translated by the developed system. There were 2,053 syllable that is being correctly translated. Thus, resulting to 91% accuracy rate. The researchers use Error Rate, Recognition Accuracy Percentage to get the accuracy formula.

Table 8

Result Record of Filipino Text to Modern Baybayin Script Translation

Trial	Total number of input	Recognized Correctly	Reliability Rate
1	100	95	100%
2	100	95	100%
3	100	95	100%

Table 4.2.1 Shows the results of the translation of Filipino Text to Modern Baybayin Script translation in terms of Reliability using Pearson R Test formula. Three (3) trials were conducted by the researchers to further test the reliability of the system. All of the trials

were having a consistent result, which result to 95 correctly translated Filipino Text. The reliability percentage is 100% based on the computation made by the researchers using the Pearson R Test Formula. See Appendix I in Appendix A, Table B to see a table that shows detailed tally of the three (3) trials conducted.

Table 9

Result Record of Filipino Text Syllable to Modern Baybayin Script Translation

Trial	Total number of input	Recognized Correctly	Reliability Rate
1	2,244	2,053	100%
2	2,244	2,053	100%
3	2,244	2,053	100%

Table 4.2.1 Shows the Result Record of Filipino Text Syllable to Modern Baybayin Script Translation in terms of Reliability. The Researchers calculated the syllables of all the testing data or the input data and it is resulted to 2,244 all in all. Researchers conducted three (3) trials to further test if the developed system is reliable enough. All trials have consistent result of 2,053 over 2,244 using the Pearson R Test Formula and it is resulted to 100% reliability since the result is consistent.

Table 10

Result Record of Modern Baybayin Script to Filipino Text Syllable Translation

Trial	Total number of input	Recognized Correctly	Reliability Rate
1	2,244	2,053	100%

2	2,244	2,053	100%
3	2,244	2,053	100%

Table 4.2.3 Shows the Result Record of Filipino Text Syllable to Modern Baybayin Script Translation in terms of Reliability. The Researchers calculated the syllables of all the testing data or the input data and it is resulted to 2,244 all in all. Researchers conducted three (3) trials to further test if the developed system is reliable enough. All trials have consistent result of 2,053 over 2,244 using the Pearson R Test Formula and it is resulted to 100% reliability since the result is consistent.

Table 11

Result Record of Modern Baybayin Script to Filipino Text Translation

Trial	Total number of input	Recognized Correctly	Reliability Rate
1	100	72	100%
2	100	72	100%
3	100	72	100%

Table 4.2.3 shows the Result Record of Modern Baybayin Script to Filipino Text Translation in terms of Reliability. There were 100 testing data that the researchers prepared to further test the reliability of the developed system. Theree (3) trials were being done and the result is consistent, there were 2 testing data translated correctly in the three trials. Thus, resulting to 100% reliability. Pearson R Test is the formula used by the researchers to test the reliability.

Table 12

Summary of Results of the System in terms of Filipino Text to Modern Baybayin

Category	Number of input	Accuracy Rate	Reliability Rate
Per Text	100	95%	100%
Per Syllable	4,250	98.6%	100%

Table 4.3.1 Shows the summary of results of the system in terms of Filipino Text to Modern Baybayin Script translation with the input of 100 set of Filipino Text.The translation per text resulted to 95% Accuracy rate since there are some text that can't be translated by the developed system. And with the reliability of 100%. As per syllable, there were 4,250 syllable in the 100 Filipino Text and resulted to 98.6% accuracy and 100% reliability.

Table 13

Summary of the Results of the System in terms of Modern Baybayin Script into Filipino Text

Category	Number of input	Accuracy Rate	Reliability Rate
Per Script	100	72%	100%
Per Syllable	2,244	91%	100%

Table 4.3.2 shows the Summary of the Results of the System in terms of Modern Baybayin Script into Filipino Text. Overall, the translation per text resulted to 72% Accuracy

rate and with the reliability of 100%. As per syllable, there were 2,244 syllable in the 100 Filipino Text and resulted to 91% accuracy rate and 100% reliability rate.

And also this study gonna interpret the answer in the Statement of the Probelm:

2. What is the significant difference in the translation of the proposed system and the existing system of Pedroza et. al?

Table 14

Comparison of proposed system to existing system of Pedroza et. al

Proposed System	Pedroza et. al System
• Can translated the syllable of C, Q, J, , V, X, Z and Ňˇ.	• Cannot translated the syllable of C, Q, J, , V, X, Z and Ňˇ
• Modern Baybayin which composed of more 17 syllables.	• Old Baybayin used which composed o f 17 syllables.
• Cannot translated the ambiguous word for example “libru” because the correct word is libro.	• Can translated ambiguous word for example “libru”.
• Can translated proper noun.	• Cannot translated proper noun.
• Can detect English word or other foreign language and not be translated.	• Cannot detect English word or other foreign language and translated.
• A two way translation of Filipino Text and Modern Baybayin	• One way translation of Filipino Text to Modern Baybayin

Table 4.3.3 shows the comparison of the proposed system made by the researcher to the existing system of Pedroza et. al. For more detailed example of words comparison of existing system make words and testing see on Appendix I, in Appendix B.

Chapter 5

SUMMARY OF FINDINGS, CONCLUSIONS AND RECCOMENDATIONS

This chapter presents the summary of all the findings of the study from gathered data conducted through experimentation. It discusses the results of the experimentation and interprets these results to come up with conclusion. Recommendations for future works will also be given based on the collected results and problems encountered in the course of the study.

5.1 Summary of Findings

Based from the summary of the findings for the translation of Filipino text to Modern Baybayin it is resulted to 95% Accuracy rate and 100% Reliability rate. As per syllable, 98.6% for the accuracy and 100% as well for the reliability because it resulted a consistent output just like the translation of Filipino Text.

In the translation of Modern Baybayin Script to Filipino text it is resulted to 72% accuracy rate and 100% reliability rate since it has a consistent result in the three trials conducted by the researchers. As per syllable, the accuracy rate reached 91% and 100% to reliability. The system shown consistency and stability in those testing being conducted, and so the developed system is proved to be accurate and reliable.

5.2 Conclusion

In this study, N-Gram and Convolutional Neural Network algorithm were being utilized to translate Filipino Text to Modern Baybayin Script and vice versa. Thorough analysis and simulation were done by the researchers to measure the accuracy and reliability of the developed system. The researchers conclude the following:

1. The researchers were able to prove that the developed system can translate Filipino Text to Modern Baybayin Script with the accuracy rate of 95%. The translation of Filipino Text Syllable to Modern Baybayin Script with the accuracy of 98.6% because the other “balbal words” cannot translated like charing, charot and cannot translated other foreign language. The researchers conclude that the developed system needs more improvement in terms of the translation of Modern Baybayin Script to Filipino Text Translation in terms of accuracy per text since the accuracy of Modern Baybayin Script to Filipino Text translation reacher 72% and 91% per syllable because there are times the system cannot validated all characters in the modern Baybayin image. The future researchers should feed more data accurately and maybe used different algorithm for the system to have a high accuracy.
2. The researchers also conclude that it already answered the statement of the problem to improve the accuracy of past research made by Pedroza et. Al with the accuracy of 99.61% and expand the scope by means of having a validation of Filipino Text, can translate Proper nouns and all kinds of letters since Modern Baybayin has 28 Script unlike the old one which is compose of 17 scripts only. It is resulted to 95% Accuracy and 98.6% reliability. See Appendix B to further see the comparison of result between the past research and this research.

5.3 Reccomendations

Based on the findings of the study, the researchers present the following recommendations for future work

1. Future researchers may develop a mobile version of this system. Because in acquired an image this prototype needed a third-party scanner to scan the image and input in the

system for the purpose of time can lessen for acquiring an image, mobile phone have captured and automatically translated the image of Modern Baybayin into text.

2. Future researchers may also improve the Modern Baybayin image to text translation in terms of having an input. They may improve in terms of materials used when writing Modern Baybayin Script.
3. Future researchers may improve the Filipino text translation in terms of automatically check the grammar of the sentence or the text being input and automatically correct it, also the spelling of the wrong Filipino words.

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APPENDICES

APPENDIX 1: INSTRUMENT

Objectives:

The following is the list of objectives for conducting the experiment paper:

- To know the accuracy of the system in terms of: (a) Filipino text to Modern Baybayin script, (b) Filipino text syllable to Modern Baybayin script, (c) Modern Baybayin script to Filipino Text Syllable, (d) Modern Baybayin to Filipino Text.
- To determine the reliability of the system's result based on the translation.
- To the significant difference in the translation of the proposed system and the existing system of Pedrozaet. al

Materials/Equipment:

1. Clean White Paper
2. Pen with black ink
3. Experimental paper
4. Digital Computer
5. Image Scanner **Procedure:**

- Filipino Text to Modern Baybayin Script translation procedure:

1. Input Filipino text in the system

2. Click submit or translate so that the inputted Filipino Text will be translated to Modern Baybayin Script.

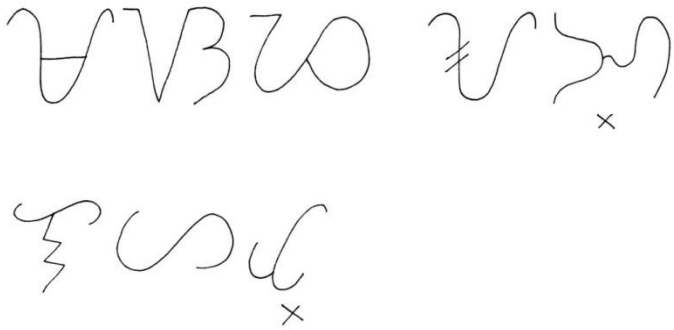
3. The output will generate.

5. Then lastly, the result of the output will be recorded to the Experimental paper.

- Modern Baybayin Script to Filipino Text translation procedure:

1. Write Modern Baybayin Script in a white clean paper using blank ink pen.

   e.g sample:



2. Scanned image to the scanner for the input in the Modern Baybayin to Filipno Text Translation.

3. Click submit or translate so that the inputted Filipino Text will be translated to Modern Baybayin Script.

4. The output will generate.

5. Then lastly, the result of the output will be recorded to the Experimental paper.

Instruction:

Table A is about the experiment paper of data for accuracy. **Table A.1** is about the Filipino Text to Modern Baybayin Script accuracy of translation, and Table **A.2**, is about the Modern Baybayin Script to Filipino Text accuracy of translation. Expert's column is the

translation made by the expert while the system column is the translation made by the system. Researchers counted the correct syllable per text and write it in the column of score per syllable and write in this column of Score per syllable while in the column correct or wrong this is for the whole text being inputted by the user, it is either 1 or 0. 1 for correct output and 0 for the wrong one.

For **Table B**, it's about the experiment paper of data for reliability. **Table B.1, it is** about the Filipino Text to Modern Baybayin Script reliability of translation, and **Table B.2,** it's about the Modern Baybayin Script to Filipino Text reliability of translation. Mark check 1 in the TP if the translation has correct output or TN if the translation have wrong output or FP if the translation have output but not correctly recognized text or FN if the output have no output but not recognized correctly the text.

Table A. Experimental Paper for Accuracy

Table 15

Filipino Text to Modern Baybayin Script

Filipino Text	Correct or Wrong	Score per Syllable	Accuracy
1. Ang bayan ko ay Malabon.	1	12/12	100%
2. Ako ay isang guro sa Paaralan ng Sekundarya ng Panghulo.	1	27/27	100%
3. Magkahawak ang ating kamay at walang kamalay malay, nakalimutan ko ang puso ko na umibig ng tunay.	1	47/47	100%
4. Ako ay nakatira sa Maysilo, bayan ng Malabon.	1	21/21	100%
5. Ang Politeknikong Unibersidad ng Pilipinas ay isa sa pinaka magandang unibersidad sa ating bansa.	1	49/49	100%
6. Ang pangalan ko ay Nilo T. Boquiren.	1	17/17	100%
7. Ang Modernong Baybayin ay dapat buhayin.	1	21/21	100%

8. May mga sariling kakilanlan na ang bawat titik sa Modernong Baybayin.	1	34/34	100%
9. Si Andrea Alovera ay isang mag-aaral sa Politeknikong Unibersidad ng Pilipinas.	1	42/42	100%
10. Pilipinas ang aking bayang sinilangan.	1	18/18	100%
11. Umakyat ako sa puno. Ako ay masaya.	1	17/17	100%
12. Si Ben ay masigasig na nag-aaral kahit na kapos sila sa pinansyal.	1	33/33	100%
13. Jose Rizal ang ngalan ng bayaning nagtanggol sa ating bayan sa pamamagitan ng pagsulat.	1	41/41	100%
14. Ang hindi magmahal sa sariling wika ay higit pa sa malansang isda.	1	32/32	100%
15. Ako ay may kaalaman ukol sa pag aaral ng Baybayin.	1	27/27	100%
16. Siya ay tumalon.	1	8/8	100%
17. Masaya ang mga bata sa kanilang mga natanggap na laruan.	1	26/26	100%
18. Ang talasalitaan ang nakatutulong para sa mga mag aaral.	1	28/28	100%
19. Pilipino at Araling Panlipunan ang aking itinuturo.	1	26/26	100%
20. Sa Puerto Prinsesa, Palawan matatagpuan ang Underground River.	1	35/35	100%
21. Siya ay nakatanggap ng papuri mula sa kanyang guro.	1	23/23	100%
22. Sina Ana, Niña at Filemon ay aking mga masisipag na estudyante.	1	32/32	100%
23. Malabon ang bayang aking kinagisnan mula pagkabata.	1	25/25	100%
24. Tila napakagandang libutin ang Pilipinas.	1	20/20	100%
25. Hala! Ang batang si Christine ay nahulog sa kanal	1	24/24	100%
26. Nakakagalak isipin na kahit papaano ay nabubuhay na ang Modernong Baybayin.	1	38/38	100%
27. Maganda si Bea.	1	7/7	100%

28. Sa ngayon ay di natin mawawari kung sino ang nagsasabi ng totoo.	1	29/29	100%
29. Di maitatangging ibang-iba na ang gawi ng mga kabataan ngayon sagawi ng kabataan noon.	1	41/41	100%
30. Ang bansang Amerika ay isa sa pinakamayamang bansa.	1	25/25	100%
31. Si Manuel L. Quezon ang nagtatag ng Quezon City.	1	25/25	100%
32. Sina Queenie at Salvi ay magkaibigang matalik.	1	25/25	100%
33. Ang akdang Romeo at Juliet ay isinulat ng isang tanyag na manunulat na si William Shakespeare.	1	50/50	100%
34. Ako na ang mangunguna upang masimulan na ang proyektong ito.	1	28/28	100%
35. Gusto kong nagbabasa ng mga salita ng Diyos.	1	20/20	100%
36. Ang Basilica Minore del Sto. Niño ay ang pinakamatandang simbahan sa Cebu.	1	35/35	100%
37. Si Melody ay kumakanta sa entablado ngayon. Siya ay nagagalak.	1	31/31	100%
38. Sina Jane at Mae ay magaling sa matematika.	1	20/20	100%
39. Sa Zamboanga nakatira ang angkan ng mga Dela Cruz.	1	24/24	100%
40. Pakiabot nga ang baso.	1	10/10	100%
41. Makikiraan po.	1	7/7	100%
42. “Manahimik kayo!”, ang wika ng guro sa kanyang mga estudyante.	1	28/28	100%
43. Asya ang pinakamalaking kontinente.	1	18/18	100%
44. Kami ay sasali sa palatuntunan sa aming paaralan. Paniguradong masaya ito.	1	35/35	100%
45. Malaki ang pinagbago ng ating bansa sa paglipas ng panahon.	1	28/28	100%
46. Ang Florante at Laura ay isinulat ni Francisco Balagtas at ang Noli Me Tangere ay isinusulat naman ni Jose Rizal.	1	58/58	100%

47. Ang Baybayin ay isang lumang paraan ng pagsulat ng mga kayumangging Pilipino.	1	38/38	100%
48. May ang ipapangalan niya sa kanyang magiging anak.	1	24/24	100%
49. Si Juan ay may maamong mukha.	1	15/15	100%
50. Si Cynthia ay naglilinis ng banyo habang ako ay nagluluto.	1	31/31	100%
51. Sa lahat ng damadamin ng puso ng tao ay wala nang mahal at dakila na gaya ng pag-ibig.	1	38/38	100%
52. Tila dambuhalang ataol ang bahay lagarian na nasasabuyan ng makinang manilaw-nilaw na liwanag.	1	46/46	100%
53. Mataas na ang araw nang lumabas si Aling Marta sa bakuran ng kanilang maliit na barong-barong.	1	44/44	100%
54. May dalawang magkaibigan na sina Felipe at Delfin.	1	25/25	100%
55. Si Don Ramon ay nakatira sa New York.	1	18/18	100%
56. Bagung-bago sa kanyang paningin ang gusali at nang isungaw niya ang kanyang mukha sa bintana ng sasakyan ay nakita niya ang mga matang nakamasid sa kanya. Isinuot niya ang salaming may kulay. Tila hindi na niya matatagalan ang nakalarawan sa mukha ng mga sumasalubong sa kanya.	1	129/129	100%
57. Umalis si Queenie sa kanilang tahanan ng umaga. Naglakad siya papuntang palengke. Nilabas niya ang kanyang pitaka. Bumili siya ng isda, karne at gulay sa palengke.	0	76/76	93%
58. Nakakahilo dumaan sa pazigzag na daan habang sakay sa jeep. Pero mas nakakahilo sumakay sa tren kaysa sa bus. Pagkatapos naming sumakay sa bus, kami ay sumakay sa barko at hindi na ako nahilo.	1	91/91	100%
59. Tuluyan nang tatakpan ng buwan ang araw at magdidilim sa kalsada. Tarantang	1	116/116	100%

magtatakbuhan ang mga tao habang nakatingala sa langit. Isang matandang babae ang takot na takot na mapapaupo sa tabi ng kalsada at magdarasal. Kakahol ang mga aso.			
60. Hoy! ang malakas niyang tawag kay Genoveva. Tara na! at tayo ay magsayawan.	0	22/35	63%
61. Magsaya! mabuhay ang mahal na birhen.	1	18/18	100%
62. Si Xavier ay naglalaro ng yoyo.	1	15/15	100%
63. Oh! Sino ang makapagsasalaysay ng mga himalang gawa ng pag-ibig.	1	31/31	100%
64. Si Donya Victorina ay nakatira sa isang apartment sa Rizal.	0	23/30	77%
65. Si Jojo ay may alagang oso.	1	13/13	100%
66. Ang maikling kwento ay isang akdang naglalathala at nagsasalaysay ng mga makatotohanang panyayari sa pang-araw-araw na buhay.	1	65/65	100%
67. Siya ay isang magandang Diyosa sa kanilang lugar	1	15/15	100%
68. Walang mang-aalipin kung walang napapaalipin.	1	22/22	100%
69. Ang oras ay ginto kaya marapat na gamitin ito mabuti.	1	26/26	100%
70. Naku! Mahuhulog ang bata.	1	11/11	100%
71. O, paliparin mo at ihalik sa Diyos, bago pa sa lupa ay sumubsob.	1	28/28	100%
72. Tanggapin mo, anak itong munting guryon na yari sa patpat at papel na mahaba.	1	37/37	100%
73. Hindi ko na alam ang gagawin sa lahat ng aking problema mga erp.Magkaka junakis na ako.	0	39/43	91%
74. Si Lakandula ay tumawa nang malakas.	1	17/17	100%
75. Siya ay isang timawa at ako naman ay isang babaylan.	1	27/27	100%
76. Nagpaplano ang Kagawaran ng Edukasyon na magkaroon ng pagbabago sa kurikulum ng mga mag-aaral. Isa sa mga pinag-iisipan nila ngayon ay kung pananatilihin pa ba o tatanggalin na ang pagbabasa ng Florante at Laura bilang bahagi ng kurikulum	1	185/185	100%

ng mga mag-aaral ngayon sa ikalawang taon. Ang paningin raw kasi ng mga mag-aaral dito ay karagdagang babasahin lamang.			
77. Ang itim na Nazareno o mas kilala rin sa Hesus na poong Nazareno ang gataong imahe ni Hesus.	1	43/43	100%
78. Bilang isang estudyanteng nagbasa ng Florante at Laura, alam ko na mayroon itong maraming aral na maari nating gamitin sa ating pang araw-araw na pamumuhay. Ang Florante at Laura ay dapat manatili sa kurikulum ng mga estudyante, dahil nilalabas nito ang kabutihan sa bawat taong nagbasa ng awit na ito. Naniniwala ako na ang mga bida sa Florante at Laura tulad ni Aladin at Florante, ay maaaring maging mabuting halimbawa sa mga mambabasang estudyante.	1	225/225	100%
79. Madaming dumating na lispu sa naganap na krime sa aming lugar sa probinsya.	1	36/36	100%
80. Tom-guts na daw ang mga bakla sabi ng kanilang ama-amahan.	0	23/28	82%
81. Ang magkakaibigang Enzo, Cedy at Valen ay magagaling sa larangan ng pagbabalita sa kanilang paaralan.	1	50/50	100%
82. Salamat sa iyong tulong mga tropa kahit nang charot ka minsan.	1	30/30	100%
83. Ang lahat ng ito ay para sa aking mga magulang na naghirap upang itaguyod ang aking pag-aaral.	1	47/47	100%
84. Magsisikap ako para sa ikagaganda ng aking buhay.	1	24/24	100%
85. Makikita mo ito sa parte noong iniligtas ni Aladdin si Florante sa mga mababangis na leon kahit magkaiba ang kanilang pinagmulan na sekta at relihiyon.	1	76/76	100%
86. Habang binabasa ko ang awit na Florante at Laura noon, napansin	1	202/202	100%

ko na maraming mga magagandang mensahe ang gustong ibigay sa atin ng makata na gumawa nito na si Francisco Balagtas. Makakabuti sa mga mag-aaral na makita nila ang mga mensahe na mayroon ang kwento na ito dahil ito ay makakatulong sa kanila na magkaroon ng magandang asal. Isa sa mga mensahe ng kwento na ito ay ang pakikiramay at pagtutulong sa kapwa.			
87. Hinding hindi tayo iiwan ni Hesus kahit anong mangyari.	1	27/27	100%
88. Walang shonget sa mundo sabi ng beki sa kanyang kaibigan.	0	21/27	78%
89. Si Jaja ang pinakamatangkad sa grupo at wala sa kanyang magkasyang sapatos.	1	36/36	100%
90. Sa lahat ng ito, naniniwala ako na uunlad muli ang bayang Pilipinas. Dahil ang kabataan ay ang pag-asa ng kinabukasan, kailangan nilang mag-aral ng mabuti, para malaman kung paano nilang lutasan ang mga problemang haharapin ng ating bansa. Ang pagtatanggal ng Florante at Laura ay hindi makakabuti sa ating bansa, at dapat ito manatili sa kurikulum ng mga estudyante sa buong Pilipinas.	1	187/187	100%
91. Magandang lalake si Jaycee kaya siya ay nagging modelo.	1	25/25	100%
92. Ang Politeknikong Unibersidad ng Pilipinas, dinadaglat bilang PUP at kilala sa mga pangalang PUP Sta. Mesa at PUP Manila, ay isang pambansang unibersidad na matatagpuan sa Santa Mesa, Maynila, Pilipinas.	1	99/99	100%
93. Masayang magdiwang ng Pista sa probinsya kasama ang erpat.	1	29/29	100%
94. Ang baybayin ay ang simbolong ginagamit ng mga sinaunang Pilipino sa pagsulat. Ito ang a ba ka da kung basahin. Ang alibata	1	125/125	100%

naman ay sinasabing nagmula sa Gitnang Silangan na mas kilala bilang alif bata. Alif ang unang letra at bata ang huli sa alpabetong iyon.			
95. Ang mga gawaing narito, na maituturing na simple at payak, ay maaaring makatulong sa iyo.	1	43/43	100%
96. Huwag kang sumuko. Maaaring ang kinabukasan mo at ng iyong pamilya ay nakasalalay sa iyong mga desisyon.	1	51/51	100%
97. Subukang alamin ang mga sakit na maaaring maidulot ng labis na pag-inom sa iyo.	1	39/39	100%
98. Sa ganitong paraan, malalaman mo ang halaga ng iyong pagbabago. Mauunawaan mo rin nang lubusan kung bakit kailangan mo nang ihinto ang pag-inom. Makikita mo nang malinaw ang halaga ng iyong pagtigil sa pag-inom at ang mga bagay na maiiwasan mo kapag naging nabago mo ang iyong buhay.	1	135/135	100%
99. Mahalaga na malaman mo na ang mga bagay na ito upang maging madali sa iyo ang pag-unawa na kailangang at dapat kang magbago sa panahon ngayon.	1	65/65	100%
100. Ang edukasyon ay kailangan ng ating mga kabataan sapagkat ito ang kanilang magiging sandata sa buhay sa kanilang kinabukasan.	1	62/62	100%
TOTAL	95	4,250/4,313	98.6%

Table 16

Modern Baybayin Script to Filipino text

No.	Modern Baybayin	Score per Syllable	Correct or Wrong	Accuracy
1	Image #1	12/12	1	100%
2	Image #2	27/27	1	100%
3	Image #3	45/47	0	96%
4	Image #4	21/21	1	100%
5	Image #5	49/49	1	100%
6	Image #6	17/17	1	88%
7	Image #7	21/21	1	88%
8	Image #8	34/34	1	100%
9	Image #9	39/42	0	93%
10	Image #10	18/18	1	100%
11	Image #11	17/17	1	100%
12	Image #12	33/33	1	100%
13	Image #13	35/41	0	85%
14	Image #14	17/32	0	53%
15	Image #15	18/27	0	67%
16	Image #16	8/8	1	100%
17	Image #17	23/26	0	88%
18	Image #18	23/28	0	82%
19	Image #19	24/26	0	88%
20	Image #20	26/35	0	75%

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21	Image #21	23/23	1	100%
22	Image #22	19	0	59%
23	Image #23	20	0	80%
24	Image #24	**20**	1	**100%**
25	Image #25	24	1	100%
26	Image #26	38	1	100%
27	Image #27	7	1	100%
28	Image #28	29	1	100%
29	Image #29	34	0	83%
30	Image #30	19	0	76%
31	Image #31	13	0	52%
32	Image #32	20	0	80%
33	Image #33	38	0	76%
34	Image #34	25	0	88%
35	Image #35	15	0	80%
36	Image #36	30	0	88%
37	Image #37	28	0	90%
38	Image #38	20	1	100%
39	Image #39	16	0	67%
40	Image #40	10	1	100%
41	Image #41	7	1	100%
42	Image #42	28	1	100%
43	Image #43	18	1	100%
44	Image #44	**35**	1	**100%**
45	Image #45	20	0	71%

46	Image #46	46	0	79%
47	Image #47	36	0	95%
48	Image #48	24	1	100%
49	Image #49	11	1	100%
50	Image #50	31	1	100%
51	Image #51	9/9	1	100%
52	Image #52	8/8	1	100%
53	Image #53	13/13	1	100%
54	Image #54	12/12	1	100%
55	Image #55	12/12	1	100%
56	Image #56	15/15	1	100%
57	Image #57	13/16	0	81%
58	Image #58	23/23	1	100%
59	Image #59	20/20	1	100%
60	Image #60	19/19	1	100%
61	Image #61	15/15	1	100%
62	Image #62	16/20	0	80%
63	Image #63	12/12	1	100%
64	Image #64	17/17	1	100%
65	Image #65	9/16	0	56%
66	Image #66	15/18	0	83%
67	Image #67	17/17	1	100%
68	Image #68	22/22	1	100%
69	Image #69	21/21	1	100%
70	Image #70	21/21	1	100%
71	Image #71	20/20	1	100%
72	Image #72	15/15	1	100%

73	Image #73	18/18	1	100%
74	Image #74	16/16	1	100%
75	Image #75	13/13	1	100%
76	Image #76	23/23	1	100%
77	Image #77	26/26	1	100%
78	Image #78	16/16	1	100%
79	Image #79	11/11	1	100%
80	Image #80	16/16	1	100%
81	Image #81	24/24	1	100%
82	Image #82	14/14	1	100%
83	Image #83	28/28	1	100%
84	Image #84	15/15	1	100%
85	Image #85	19/19	1	100%
86	Image #86	23/23	1	100%
87	Image #87	17/17	1	100%
88	Image #88	14/14	1	100%
89	Image #89	16/16	1	100%
90	Image #90	17/17	1	100%
91	Image #91	15/15	1	100%
92	Image #92	16/16	1	100%
93	Image #93	19/19	1	100%
94	Image #94	17/17	1	100%
95	Image #95	20/20	1	100%
96	Image #96	17/17	1	100%
97	Image #97	17/17	1	100%
98	Image #98	13/13	1	100%
99	Image #99	19/19	1	100%

100	Image #100	20/20	1	100%

Table B. Experimental Paper for Reliability

Table 17

Filipino Text to Modern Baybayin Script

No.	Filipino Text	Correct or Wrong		
		Trial 1	Trial 2	Trial 3
1	Ang bayan ko ay Malabon.	1	1	1
2	Ako ay isang guro sa Paaralan ng Sekundarya ng Panghulo.	1	1	1
3	Magkahawak ang ating kamay at walang kamalay malay, nakalimutan ko ang puso ko na umibig ng tunay.	1	1	1
4	Ako ay nakatira sa Maysilo, bayan ng Malabon.	1	1	1
5	Ang Politeknikong Unibersidad ng Pilipinas ay isa sa pinaka magandang unibersidad sa ating bansa.	1	1	1
6	Ang pangalan ko ay Nilo T. Boquiren.	1	1	1
7	Ang Modernong Baybayin ay dapat buhayin.	1	1	1
8	May mga sariling kakilanlan na ang bawat titik sa Modernong Baybayin.	1	1	1
9	Si Andrea Alovera ay isang mag-aaral sa Politeknikong Unibersidad ng Pilipinas.	1	1	1
10	Pilipinas ang aking bayang sinilangan.	1	1	1
11	Umakyat ako sa puno. Ako ay masaya.	1	1	1
12	Si Ben ay masigasig na nag-aaral kahit na kapos sila sa pinansyal.	1	1	1
13	Jose Rizal ang ngalan ng bayaning nagtanggol sa ating bayan sa pamamagitan ng pagsulat.	1	1	1
14	Ang hindi magmahal sa sariling wika ay higit pa sa malansang isda.	1	1	1
15	Ako ay may kaalaman ukol sa pag aaral ng Baybayin.	1	1	1

16	Siya ay tumalon.	1	1	1
17	Masaya ang mga bata sa kanilang mga natanggap na laruan.	1	1	1
18	Ang talasalitaan ang nakatutulong para sa mga mag aaral.	1	1	1
19	Pilipino at Araling Panlipunan ang aking itinuturo.	1	1	1
20	Sa Puerto Prinsesa, Palawan matatagpuan ang Underground River.	1	1	1
21	Siya ay nakatanggap ng papuri mula sa kanyang guro.	1	1	1
22	Sina Ana, Niña at Filemon ay aking mga masisipag na estudyante.	1	1	1
23	Malabon ang bayang aking kinagisnan mula pagkabata.	1	1	1
24	Tila napakagandang libutin ang Pilipinas.	1	1	1
25	Hala! Ang batang si Christine ay nahulog sa kanal	1	1	1
26	Nakakagalak isipin na kahit papaano ay nabubuhay na ang Modernong Baybayin.	1	1	1
27	Maganda si Bea.	1	1	1
28	Sa ngayon ay di natin mawawari kung sino ang nagsasabi ng totoo.	1	1	1
29	Di maitatangging ibang-iba na ang gawi ng mga kabataan ngayon sagawi ng kabataan noon.	1	1	1
30	Ang bansang Amerika ay isa sa pinakamayamang bansa.	1	1	1
31	Si Manuel L. Quezon ang nagtatag ng Quezon City.	1	1	1
32	Sina Queenie at Salvi ay magkaibigang matalik.	1	1	1
33	Ang akdang Romeo at Juliet ay isinulat ng isang tanyag na manunulat na si William Shakespeare.	1	1	1
34	Ako na ang mangunguna upang masimulan na ang proyektong ito.	1	1	1
35	Gusto kong nagbabasa ng mga salita ng Diyos.	1	1	1
36	Ang Basilica Minore del Sto. Niño ay ang pinakamatandang simbahan sa Cebu.	1	1	1
37	Si Melody ay kumakanta sa entablado ngayon. Siya ay nagagalak.	1	1	1
38	Sina Jane at Mae ay magaling sa matematika.	1	1	1

39	Sa Zamboanga nakatira ang angkan ng mga Dela Cruz.	1	1	1
40	Pakiabot nga ang baso.	1	1	1
41	Makikiraan po.	1	1	1
42	“Manahimik kayo!”, ang wika ng guro sa kanyang mga estudyante.	1	1	1
43	Asya ang pinakamalaking kontinente.	1	1	1
44	Kami ay sasali sa palatuntunan sa aming paaralan. Paniguradong masaya ito.	1	1	1
45	Malaki ang pinagbago ng ating bansa sa paglipas ng panahon.	1	1	1
46	Ang Florante at Laura ay isinulat ni Francisco Balagtas at ang Noli Me Tangere ay isinusulat naman ni Jose Rizal.	1	1	1
47	Ang Baybayin ay isang lumang paraan ng pagsulat ng mga kayumangging Pilipino.	1	1	1
48	May ang ipapangalan niya sa kanyang magiging anak.	1	1	1
49	Si Juan ay may maamong mukha.	1	1	1
50	Si Cynthia ay naglilinis ng banyo habang ako ay nagluluto.	1	1	1
51	Sa lahat ng damadamin ng puso ng tao ay wala nang mahal at dakila na gaya ng pag-ibig.	1	1	1
52	Tila dambuhalang ataol ang bahay lagarian na nasasabuyan ng makinang manilaw-nilaw na liwanag.	1	1	1
53	Mataas na ang araw nang lumabas si Aling Marta sa bakuran ng kanilang maliit na barong-barong.	1	1	1
54	May dalawang magkaibigan na sina Felipe at Delfin.	1	1	1
55	Si Don Ramon ay nakatira sa New York.	1	1	1
56	Bagung-bago sa kanyang paningin ang gusali at nang isungaw niya ang kanyang mukha sa bintana ng sasakyan ay nakita niya ang mga matang nakamasid sa kanya. Isinuot niya ang salaming may kulay. Tila hindi na niya matatagalan ang nakalarawan sa mukha ng mga sumasalubong sa kanya.	1	1	1
57	Umalis si Queenie sa kanilang tahanan ng umaga. Naglakad siya papuntang palengke. Nilabas niya ang kanyang pitaka. Bumili siya ng isda, karne at gulay sa palengke.	0	0	0

58	Nakakahilo dumaan sa pazigzag na daan habang sakay sa jeep. Pero mas nakakahilo sumakay sa tren kaysa sa bus. Pagkatapos naming sumakay sa bus, kami ay sumakay sa barko at hindi na ako nahilo.	1	1	1
59	Tuluyan nang tatakpan ng buwan ang araw at magdidilim sa kalsada. Tarantang magtatakbuhan ang mga tao habang nakatingala sa langit. Isang matandang babae ang takot na takot na mapapaupo sa tabi ng kalsada at magdarasal. Kakahol ang mga aso.	1	1	1
60	Hoy! ang malakas niyang tawag kay Genoveva. Tara na! at tayo ay magsayawan.	0	0	0
61	Magsaya! mabuhay ang mahal na birhen.	1	1	1
62	Si Xavier ay naglalaro ng yoyo.	1	1	1
63	Oh! Sino ang makapagsasalaysay ng mga himalang gawa ng pag-ibig.	1	1	1
64	Si Donya Victorina ay nakatira sa isang apartment sa Rizal.	0	0	0
65	Si Jojo ay may alagang oso.	1	1	1
66	Ang maikling kwento ay isang akdang naglalathala at nagsasalaysay ng mga makatotohanang panyayari sa pang-araw-araw na buhay.	1	1	1
67	Siya ay isang magandang Diyosa sa kanilang lugar	1	1	1
68	Walang mang-aalipin kung walang napapaalipin.	1	1	1
69	Ang oras ay ginto kaya marapat na gamitin ito mabuti.	1	1	1
70	Naku! Mahuhulog ang bata.	1	1	1
71	O, paliparin mo at ihalik sa Diyos, bago pa sa lupa ay sumubsob.	1	1	1
72	Tanggapin mo, anak itong munting guryon na yari sa patpat at papel na mahaba.	1	1	1
73	Hindi ko na alam ang gagawin sa lahat ng aking problema mga erp.Magkaka junakis na ako.	0	0	0
74	Si Lakandula ay tumawa nang malakas.	1	1	1
75	Siya ay isang timawa at ako naman ay isang babaylan.	1	1	1
76	Nagpaplano ang Kagawaran ng Edukasyon na magkaroon ng pagbabago sa kurikulum ng mga mag-aaral. Isa sa mga pinag-iisipan nila	1	1	1

	ngayon ay kung pananatilihin pa ba o tatanggalin na ang pagbabasa ng Florante at Laura bilang bahagi ng kurikulum ng mga mag-aaral ngayon sa ikalawang taon. Ang paningin raw kasi ng mga mag-aaral dito ay karagdagang babasahin lamang.			
77	Ang itim na Nazareno o mas kilala rin sa Hesus na poong Nazareno ang gataong imahe ni Hesus.	1	1	1
78	Bilang isang estudyanteng nagbasa ng Florante at Laura, alam ko na mayroon itong maraming aral na maari nating gamitin sa ating pang araw-araw na pamumuhay. Ang Florante at Laura ay dapat manatili sa kurikulum ng mga estudyante, dahil nilalabas nito ang kabutihan sa bawat taong nagbasa ng awit na ito. Naniniwala ako na ang mga bida sa Florante at Laura tulad ni Aladin at Florante, ay maaaring maging mabuting halimbawa sa mga mambabasang estudyante.	1	1	1
79	Madaming dumating na lispu sa naganap na krime sa aming lugar sa probinsya.	1	1	1
80	Tom-guts na daw ang mga bakla sabi ng kanilang ama-amahan.	1	1	1
81	Ang magkakaibigang Enzo, Cedy at Valen ay magagaling sa larangan ng pagbabalita sa kanilang paaralan.	1	1	1
82	Salamat sa iyong tulong mga tropa kahit nang charot ka minsan.	1	1	1
83	Ang lahat ng ito ay para sa aking mga magulang na naghirap upang itaguyod ang aking pag-aaral.	1	1	1
84	Magsisikap ako para sa ikagaganda ng aking buhay.	1	1	1
85	Makikita mo ito sa parte noong iniligtas ni Aladdin si Florante sa mga mababangis na leon kahit magkaiba ang kanilang pinagmulan na sekta at relihiyon.	1	1	1
86	Habang binabasa ko ang awit na Florante at Laura noon, napansin ko na maraming mga magagandang mensahe ang gustong ibigay sa atin ng makata na gumawa nito na si Francisco Balagtas. Makakabuti sa mga mag-aaral na makita nila ang mga mensahe na mayroon ang kwento na ito dahil ito ay makakatulong sa kanila na magkaroon ng magandang asal. Isa sa mga mensahe ng kwento na ito ay ang pakikiramay at pagtutulong sa kapwa.	1	1	1

87	Hinding hindi tayo iiwan ni Hesus kahit anong mangyari.	1	1	1
88	Walang shonget sa mundo sabi ng beki sa kanyang kaibigan.	1	1	1
89	Si Jaja ang pinakamatangkad sa grupo at wala sa kanyang magkasyang sapatos.	1	1	1
90	Sa lahat ng ito, naniniwala ako na uunlad muli ang bayang Pilipinas. Dahil ang kabataan ay ang pag-asa ng kinabukasan, kailangan nilang mag-aral ng mabuti, para malaman kung paano nilang lutasan ang mga problemang haharapin ng ating bansa. Ang pagtatanggal ng Florante at Laura ay hindi makakabuti sa ating bansa, at dapat ito manatili sa kurikulum ng mga estudyante sa buong Pilipinas.	1	1	1
91	Magandang lalake si Jaycee kaya siya ay nagging modelo.	1	1	1
92	Ang Politeknikong Unibersidad ng Pilipinas, dinadaglat bilang PUP at kilala sa mga pangalang PUP Sta. Mesa at PUP Manila, ay isang pambansang unibersidad na matatagpuan sa Santa Mesa, Maynila, Pilipinas.	1	1	1
93	Masayang magdiwang ng Pista sa probinsya kasama ang erpat.	1	1	1
94	Ang baybayin ay ang simbolong ginagamit ng mga sinaunang Pilipino sa pagsulat. Ito ang a ba ka da kung basahin. Ang alibata naman ay sinasabing nagmula sa Gitnang Silangan na mas kilala bilang alif bata. Alif ang unang letra at bata ang huli sa alpabetong iyon.	1	1	1
95	Ang mga gawaing narito, na maituturing na simple at payak, ay maaaring makatulong sa iyo.	1	1	1
96	Huwag kang sumuko. Maaaring ang kinabukasan mo at ng iyong pamilya ay nakasalalay sa iyong mga desisyon.	1	1	1
97	Subukang alamin ang mga sakit na maaaring maidulot ng labis na pag-inom sa iyo.	1	1	1
98	Sa ganitong paraan, malalaman mo ang halaga ng iyong pagbabago. Mauunawaan mo rin nang lubusan kung bakit kailangan mo nang ihinto ang pag-inom. Makikita mo nang malinaw ang halaga ng iyong pagtigil sa pag-inom at ang mga bagay na maiiwasan mo kapag naging nabago mo ang iyong buhay.	1	1	1
99	Mahalaga na malaman mo na ang mga bagay na ito upang maging madali sa iyo ang pag-	1	1	1

	unawa na kailangang at dapat kang magbago sa panahon ngayon.			
100	Ang edukasyon ay kailangan ng ating mga kabataan sapagkat ito ang kanilang magiging sandata sa buhay sa kanilang kinabukasan.	1	1	1

Table 18

Modern Baybayin Script to Filipino Text

No.	Filipino Text	Correct or Wrong		
		Trial 1	Trial 2	Trial 3
1	Image #1	1	1	1
2	Image #2	1	1	1
3	Image #3	1	1	1
4	Image #4	1	1	1
5	Image #5	1	1	1
6	Image #6	1	1	1
7	Image #7	1	1	1
8	Image #8	1	1	1
9	Image #9	1	1	1
10	Image #10	1	1	1
11	Image #11	1	1	1
12	Image #12	1	1	1
13	Image #13	1	1	1
14	Image #14	1	1	1
15	Image #15	1	1	1
16	Image #16	1	1	1
17	Image #17	1	1	1
18	Image #18	1	1	1

19	Image #19	1	1	1
20	Image #20	1	1	1
21	Image #21	1	1	1
22	Image #22	0	0	0
23	Image #23	0	0	0
24	Image #24	1	1	1
25	Image #25	1	1	1
26	Image #26	1	1	1
27	Image #27	1	1	1
28	Image #28	1	1	1
29	Image #29	0	0	0
30	Image #30	0	0	0
31	Image #31	0	0	0
32	Image #32	0	0	0
33	Image #33	0	0	0
34	Image #34	0	0	0
35	Image #35	0	0	0
36	Image #36	0	0	0
37	Image #37	0	0	0
38	Image #38	1	1	1
39	Image #39	0	0	0
40	Image #40	1	1	1
41	Image #41	1	1	1
42	Image #42	1	1	1
43	Image #43	1	1	1
44	Image #44	1	1	1

45	Image #45	0	0	0
46	Image #46	0	0	0
47	Image #47	0	0	0
48	Image #48	1	1	1
49	Image #49	1	1	1
50	Image #50	1	1	1
51	Image #51	1	1	1
52	Image #52	1	1	1
53	Image #53	1	1	1
54	Image #54	1	1	1
55	Image #55	1	1	1
56	Image #56	1	1	1
57	Image #57	0	0	0
58	Image #58	1	1	1
59	Image #59	1	1	1
60	Image #60	1	1	1
61	Image #61	1	1	1
62	Image #62	0	0	0
63	Image #63	1	1	1
64	Image #64	1	1	1
65	Image #65	0	0	0
66	Image #66	0	0	0
67	Image #67	1	1	1
68	Image #68	1	1	1
69	Image #69	1	1	1
70	Image #70	1	1	1

71	Image #71	1	1	1
72	Image #72	1	1	1
73	Image #73	1	1	1
74	Image #74	1	1	1
75	Image #75	1	1	1
76	Image #76	1	1	1
77	Image #77	1	1	1
78	Image #78	1	1	1
79	Image #79	1	1	1
80	Image #80	1	1	1
81	Image #81	1	1	1
82	Image #82	1	1	1
83	Image #83	1	1	1
84	Image #84	1	1	1
85	Image #85	1	1	1
86	Image #86	1	1	1
87	Image #87	1	1	1
88	Image #88	1	1	1
89	Image #89	1	1	1
90	Image #90	1	1	1
91	Image #91	1	1	1
92	Image #92	1	1	1
93	Image #93	1	1	1
94	Image #94	1	1	1
95	Image #95	1	1	1
96	Image #96	1	1	1

97	Image #97	1	1	1
98	Image #98	1	1	1
99	Image #99	1	1	1
100	Image #100	1	1	1

Text	Correct or Wrong	
	Previous Research	Our Research
nabaak	1	1
bababa	1	1
babaha	1	1
babakas	1	1
nabakbak	1	1
nabakuna	1	1
mabalat	1	1
nabaligtad	1	1
babalik	1	1
nabalot	1	1
babanat	1	1
nagbanlaw	1	1
nagbantay	1	1
nagbaon	1	1
babaril	1	1
nabasa	1	1
nabasag	1	1
nagdabog	1	1
nadakot	1	1
nadala	1	1
dadalaw	1	1
dadaldal	1	1
dadaloy	1	1
madami	1	1
madamot	1	1
magdasal	1	1
nadating	1	1
gagabay	1	1
nagapang	1	1
nagalaw	1	1
nagalit	1	1
nagastos	1	1
nagawa	1	1
magaya	1	1
magising	1	1
guguhit	1	1
guguho	1	1
magulang	1	1
magulat	1	1
nagunting	1	1
nagusot	1	1
magutom	1	1
hahabol	1	1
nahagis	1	1
hahakbang	1	1
hahalik	1	1
nahalo	1	1
mahila	1	1
iigib	1	1
i-zigzag	0	1
nagkalabog	1	1
nakalantog	1	1
nakamot	1	1
kakanta	1	1
kikilig	1	1

kikilos	1	1
kokopya	1	1
makulay	1	1
lalabas	1	1
nilagay	1	1
malakas	1	1
lalapit	1	1
lilibot	1	1
lilikha	1	1
malinis	1	1
malukot	1	1
magluto	1	1
magnakaw	1	1
nangisi	1	1
nangiti	1	1
mala-doña	1	1
napilit	1	1
napisa	1	1
mapitas	1	1
paki-zipper	0	0
pupulot	1	1
pupuna	1	1
pupuno	1	1
masabi	1	1
sasabit	1	1
sasabog	1	1
sasaboy	1	1
nasadya	1	1
sasakay	1	1
masakit	1	1
magsalamin	1	1
sasalin	1	1
sasalubong	1	1
sasalungat	1	1
sasampay	1	1
masanay	1	1
nasanggi	1	1
masapal	1	1
pinagsumbahan	1	1
nasikap	1	1
magsilbi	1	1
masindak	1	1
nasingil	1	1
sisipa	1	1
masira	1	1
pasikat	1	1
sisiyasat	1	1
masolo	1	1
bumiñan	0	1
cuadrahan	0	1

APPENDIX 2: CORRESPONDENCE

Mr. Nilo T. Boquiren a Teacher I from Pnaghulo National High School who teach Araling Panlipunan and Filipino Subject to Junior High School Student for almost 5 years. He has a knowledge in expertise in Baybayin.

APPENDIX 3: CERTIFICATION OF EDITING

CERTIFICATION OF EDITING

This is to certify that the thesis work entitled Filipino Text to Modern Baybayin Script in Image Form Using N-Gram & Modern Baybayin Script in Image Form to Filipino Text Using Convolutional Neural Network by Alovera, Andrea B., Daisog, Greg Daniel B., Escarda, Jericho G., and Gonzales, Benedick G. was proofread and edited by the undersigned.

This certification is being issued for whatever legal purpose it may serve.

Signed: Christine L. Aquino

Date: March 31, 2019

APPENDIX 4: BIOGRAPHICAL STATEMENT

He studied primary education in Talolong Elementary School till his 4th Grade(2004-2009) and transferred in Manila in his 5th Grade. He continued his primary education in Santolan Elementary School for 2 years(2009-20011) and studied secondary education in Santolan High School for 4 years(2011-2015). He is a 4th year student and currently studying in Polytechnic University of the Philippines Academic Year 2018-2019 with a Bachelor Degree in Computer Science. He is currently living with his parents and 3 brothers in Santolan, Pasig City. He has also scholarship which aid his other financial problem. He is a Baptist and the church he is attending is the Sumulong Baptist Church. He is an introvert person because he doesn't want to talk as much as possible, the only time he wants to talk is when there is something important to talk about.

Benedick G. Gonzales is the son of Helen G. Gonzales and Johnson R. Gonzales. His two brothers were John Patrick G. Gonzales and Jonel G. Gonzales and two sister named Jonnalyn G. Gonzales and Michelle G. Gonzales. He's a good child of his parent and he is intelligent and industrious. He studied in Polytechnic University of the Philippines and he takes a course Computer Science. In elementary he has a good honor on his study in Burgos Elementary school and joining in any drawing contest and in secondary he has been a SSG officer in Burgos National High School and a journalist and a cartoonist in The Greenstalk and have been joining in competition in Araling Panlipunan.

A graduate of Dr. Manuel S. Tansinsin Memorial School in in his elementary education and of Assumpta Academy in high school years, Greg Daniel Nierves Daisog is currently a graduating student of the Polytechnic University of the Philippines batch 2018-2019 with the course of Bachelor of Science in Computer Science with the specialization in Information Systems Development. A middle child among three siblings and has a knack for universal history and archaeology. An aspiring back-end web developer and COBOL programmer, he is pretty much excited to learn more about the IT industry and expand his technical knowledge in the world of programs.

Andrea B. Alovera developed an early interest in computer when back when she was in high school. She spent his High School years at Jeremiah Christian Academy for four (4) years. She's a graduating student of Polytechnic University of the Philippines Sta., Mesa Manila with the degree of Bachelor of Science in Computer Science with specialization to Information System Development. During her college years, She is involved in 2 organizations: Association of Student for Computer Intelligence and Integration and PUP College Y-Club affiliate by YMCA Manila. She is also knowledgeable in different programming language such as Java, COBOL, C and etc but her most favorite is Web programming language.

APPENDIX 5: IMPLEMENTATION REPORT

Introduction:

A BaybayVersa is a Desktop application which translates Filipino Text to Modern Baybayin Script using N-Gram and Modern Baybayin Script to Filipino Text using Convolutional Neural Network or Image Processing. The input in terms of Filipino Text is the user should type it in the text box in the user interface while the input in Modern Baybayin Script should be in Image form. The user hould capture tha Baybayin Script using their mobile phones then transfer it in the desktop or laptop then input in the system. The researchers made a study that convert Filipino Text into Modern Baybayin Script a bi-directional translation of Modern Baybayin Script into Filipino Text. Modern Baybayin Characters is composed of 28 base characters which says that it can translate proper nouns because all the alphabet has equivalent to Modern Baybayin Characters.

The aim of the developed system has been meet by having a high accuracy and high reliability result.

Statement of the Problem:

This study aims to solve the problem about translating proper nouns of the existing system by Pedroza et. al and to increase the degree of accuracy using convolutional neural network in correctly recognizing modern baybayin characters. In addition, the researchers intend to answer the following questions at the end of the study:

1. What is the degree of accuracy of translation with respect to:

    a. Filipino Text to Modern Baybayin script?

    b. Filipino Text Syllable to Modern Baybayin Script?

c. Modern Baybayin Script to Filipino Text Syllable?

d. Modern Baybayin script to Filipino text?

2. What is the significant difference in the translation of the proposed system and the existing system of Pedrozaet. al?

Respondents

The training data of this developed system came from different students of Polytechnic University of the Philippines. It is conducted by the researchers to get the different strokes of Modern Baybayin of different people since the Modern Baybayin Script translation need an image to be translated to Filipino Text.

Proportional and non-proportional quota sampling was used by the researchers to select data. Proportional and non-proportional quota sampling, members are sampled until exact proportions of certain types of data are obtained or until sufficient data in different categories is collected (CIRT, n.d.). Using Proportional and non-proportional quota sampling, the researchers prepared data for text to image translation and vice versa.

Implementation Period Time Frame

Experimentation and calculation were the two parts used by the researchers to develop this system. Experimentation, the researchers having a hard time to increase the accuracy of the developed system that’s why a lots of data gathering were conducted time to time to meet the right accuracy and reliability of the system. The data gathering was made during the development of the tool in the mid November to December. The table below shows the time frame during the researchers developed this research.

	JANUARY		FEBRUARY				MARCH	
	2ND Week	**4th Week**	**1st Week**	**2nd Week**	**3rd Week**	**4th Week**	**1st Week-2nd Week**	**3rd Week – 4th Week**
1. Continuous Data Gathering								
2. Developing the tool								
3. Cleaning the tool								
4. Testing the Tool								
5. Fixing the bugs in the tool								
6. Improving the accuracy of the tool								
7. Experiment Paper								
8. Finalization of the Documentation								

Implementation Procedures

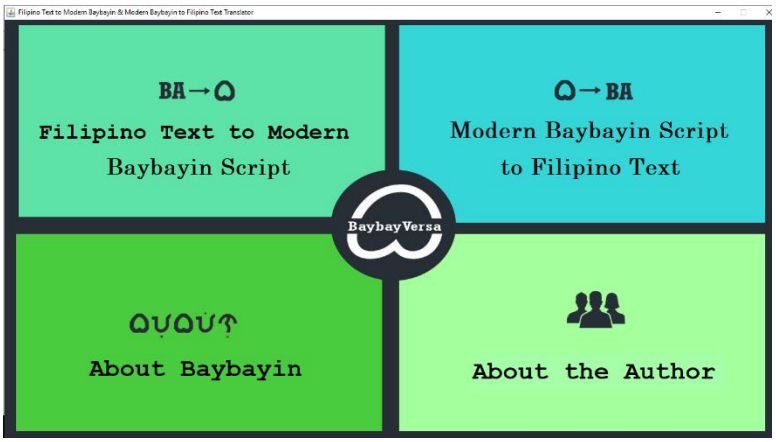
The implementation procedure of BaybayVersa was achieved through different techniques. The researchers have the accuracy and reliability formula to further test if the developed system meets its statement of the problem. (See Appendix I) To further see the details. There were 100 sets of Filipino Text and Modern Baybayin Script in Image Form to be translated by the developed system. Three (3) trials were conducted to further test its reliability.

The implementation procedure of Baybay Versa was achieved through different procedures. After developing the tool, the researchers had the testing for the accuracy and reliability of the system in order to satisfy the following statement of the problem: (1)

What is the degree of accuracy of translation with respect to: a. Filipino Text to Modern Baybayin script?, B. Filipino Text Syllable to Modern Baybayin Script?, c. Modern Baybayin Script to Filipino Text Syllable?, and d. Modern Baybayin script to Filipino text? (2) What is the significant difference in the translation of the proposed system and the existing system of Pedrozaet. al? For the testing of the system, a total of 100 testing data was gathered from a selected pool of respondets. This data was then presented to the expert of the system, Mr. Nilo Boqueren whom in return altered, finalized and approved the given set of data. Once approved the researchers utilized the approved data to conduct the actual testing of the tool in order to get the corresponding accuracy and this testing was done three times in order to obtain the reliability.

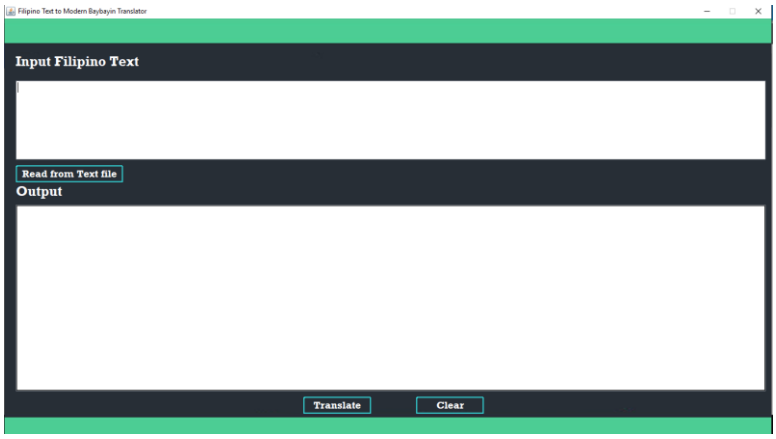
Screenshots of the System

Figure 17. First User Interface Window



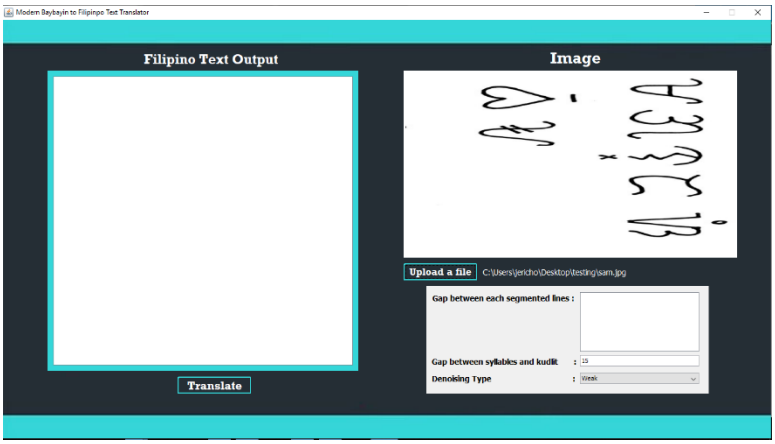
“BAYBAY VERSA” a name of the developed system and it is place in the center of the interface. The interface has a 4 buttons such as Filipino Text to Modern Baybayin Script, Modern Baybayin Script to Filipino Text, About Baybayin and About the Author. Every buttons has a corresponding actions like when the user clicked the About the Author, the developer of the system will viewed.

Figure 18. Filipino Text to Modern Baybayin Script Window



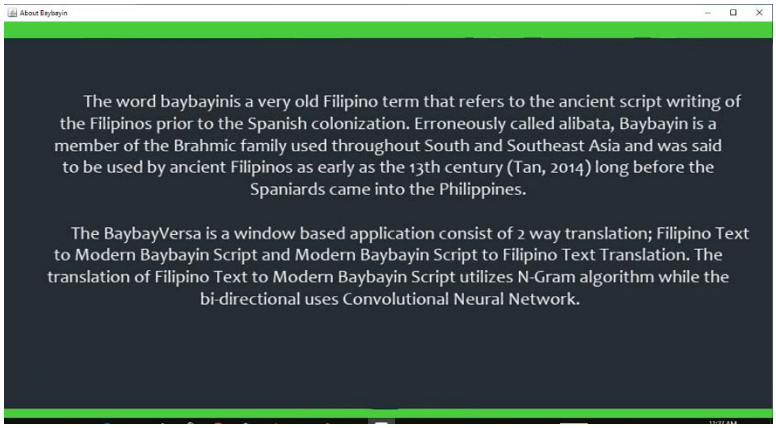
When the user input Filipino Text, phrase or sentence by typing in text field or importing a text file and the output gonna display by clicking the button translate. Clear button to clear all the output and the input of the user.

Figure 19. Modern Baybayin Script to Filipino Text Window



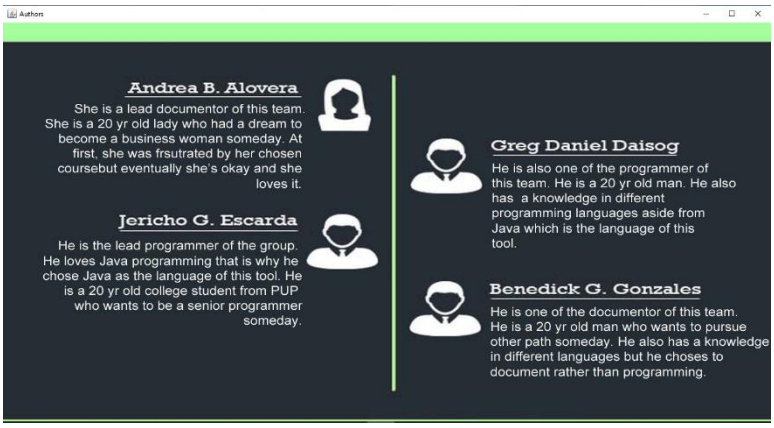
When the user input an image of Modern baybayin through the use of black pen drawing in white bond paper and gonna scan by any scanner devices or application and user gonna upload the picture of the Modern Baybayin and pick it in his/her directory folder and click the translate button to see the corresponding Filipino Text equivalent.

Figure 20. About Baybayin Window



A brief description and insight about Baybayin for the user will have some learnings about it.

Figure 21. About the Author Window



A brief description about the developer of this system and a brief insight about their own life.